Multiview SAR Target Recognition Based on Adaptive Semantic-View Feature Embedding Fusion Network

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***Abstract*—In synthetic aperture radar (SAR) automatic target recognition (ATR) tasks, due to the SAR imaging mechanism and the characteristics of target scattering structures, the image features of a target under a single observation angle may be insufficiently expressed, and the image features under different observation angles may vary significantly, which makes target recognition using single-view SAR images challenging. Therefore, acquiring multiview images of the target for joint recognition has become a promising method. However, existing multiview recognition models suffer from problems, such as insufficient learning of multiview features, inapplicability under incomplete multiview conditions, and scarcity of multiview training data, restricting the performance and practicality. In this article, we establish a new multiview SAR target recognition framework to uniformly address the above issues. First, we construct a semantic-view dual-encoder feature embedding fusion network (SeVi). By performing classification modeling on the observation views of images to introduce view prior information, the model is guided to acquire and fuse view-aware features. Second, we propose a prototype-based missing view completion strategy, enabling the model to adapt to complex incomplete multiview scenarios while maintaining high performance. Third, we put forward a nonstrict multiview recognition data construction method based on semi-supervised classification, which efficiently acquires data to support the training of multiview recognition models. Experiments are conducted on multiple vehicle and aircraft target recognition datasets, including ATRNet-STAR, MSTAR, and SARAircraft. The results demonstrate that the proposed model can effectively capture and fuse view-aware features, achieving leading performance compared with existing SAR multiview target recognition methods. Moreover, it maintains high performance even under incomplete multiview complex conditions, which verifies the robustness and flexibility of the proposed multiview recognition framework.**

***Index Terms*—Feature embedding, fusion network, multiview, synthetic aperture radar (SAR), target recognition.**

I. Introduction

IN RECENT years, the rapid advancements of deep learning computer vision technologies and high-resolution synthetic aperture radar (SAR) imaging techniques have greatly accelerated the development of SAR automatic target recognition (ATR) research. For targets, such as ships, aircrafts, and vehicles, cutting-edge SAR ATR research has advanced from basic classification tasks to tasks, such as few-shot recognition, cross-domain recognition, and multimodal recognition. A large number of excellent works have emerged in aspects, such as deep learning-based models and datasets [1], [2], [3], [4], [5], [6]. However, due to challenges such as the difficulty of data acquisition on SAR remote sensing platforms and the low cooperation level between the platform and the targets of interest, the current research mainly focuses on recognition from a single observation angle, with limited use of multiview observation images for joint target recognition [7]. The single observation configuration between the SAR platform and the targets of interest leads to two main problems: 1) under a single view, the image features of the targets may not fully capture the intrinsic characteristics of the targets and 2) from different views, the differences between different image features of the same target may be significant. As a result, in scenarios with these issues, the performance of target recognition using only a single-view image may be limited. The essence of SAR imaging is the measurement of the backscattering coefficient of ground objects [8]. The features of SAR target images are mainly determined by their strong scattering structures, such as the rough high-reflectivity surfaces of targets, the dihedral angles formed by the targets themselves or between the targets and the environment, etc. [9]. When radar waves illuminate the target from different views, the energy reflected by different parts of the target varies, leading to significant changes in the image features. The worst thing is that the features of the target image may not be obvious from a specific view, making the target unrecognizable. In practical tasks, it is difficult to preset a good observation angle for the target to avoid this situation. Therefore, it is expected to more comprehensively capture the features of the targets by observing the target from multiple views and performing multiview information fusion processing, thereby enhancing recognition performance.

The definitions for the view in remote sensing observations can be various [10]. In a broad sense, a *view* generally refers to an observation that contains information about the target of interest; therefore, multiview can involve multiple sensors, multiple temporal phases, multiple wavebands, or multiple

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modalities [11]. In this article, a *view* is specifically defined as the observation angle of the sensor relative to the target in the target coordinate system. Multiview images of a target can be obtained by the same sensor at different positions through its relative motion with respect to the target, or they can be acquired simultaneously by multiple distributed sensors. We focus more on platforms with identical radar systems, as the emphasis lies on the inherent differences of targets in images from different views, rather than domain biases caused by factors other than viewing angles.

The multiview target recognition task is to construct a classification model. The input is images of the same target from multiple different views, and the output is the semantic category of that target. A deep learning model can perform feature-level fusion on images from multiple views. Compared with the traditional method of first recognizing each single-view image and then conducting decision-level fusion on the results, the more fine-grained fusion brought by a multiview deep learning model is expected to yield more accurate recognition results [7].

Multiview fusion, like other remote sensing vision tasks, such as multispectral fusion and multiscale fusion, essentially involves leveraging the correlation and complementarity between multiple images of a target to perform downstream tasks [12], [13], [14]. In the multiview fusion tasks, correlation refers to the association of the same feature of the target across different views, while complementarity means that a single view can only capture part of the features of the target, and features that cannot be obtained from one view may be acquired from another view. The specialty of multiview fusion lies in the fact that multiview images are nonaligned or nonregistration, making pixel-level processing impossible. Instead, it is necessary to transform the spatial features into high-dimensional semantic features. During the multiview image acquisition process, factors such as differences and variations in sensor detection capabilities, changes in the state of the target, and changes in environmental conditions all affect the acquisition of target features, posing challenges for target recognition based on multiview fusion [15].

The basic approach of existing multiview feature fusion methods is to first construct a feature extraction module based on convolutional neural networks (CNNs) to extract multiview features. Then, a feature fusion module based on CNNs, multilayer perceptrons (MLPs), long short-term memory (LSTM) networks, etc., is built to obtain fused features, followed by the use of the fused features to complete classification tasks [16]. This architecture typically relies on an assumption that the viewpoint information of multiple images used to identify a target is a priori. In other words, the observation angle of the image corresponding to each view of the target is predefined, and the order in which the images are input into the model is fixed. However, when the view of the target image is unknown or shuffled, the model may not function as expected. To address this issue, we propose the idea of introducing view-aware prior information [17], allowing the model to extract view features from the images and better guide the multiview feature fusion, with more relaxed input requirements. While it is difficult to directly label which areas in a particular view are beneficial for target recognition, it is easy to classify the view of the image based on prior knowledge of the structure of the target and assign one-hot view labels. Therefore, we propose a semantic-view dual-branch embedding fusion network named *SeVi*, which consists of three main parts: 1) a dual-branch feature extraction module based on CNNs that can extract semantic feature maps and view feature maps for each view image, with the semantic feature map undergoing view-aware embedding; 2) a multiview feature fusion module based on a Transformer network, which uses the multihead self-attention mechanism to perform feature-level fusion of multiview embeddings; and 3) a hybrid loss function based on multitask learning, which jointly optimizes semantic feature extraction, view feature extraction, and multiview semantic classification.

A critical challenge confronting existing multiview recognition models is the potential absence of partial view images for target samples [18]. In practical tasks, training data may contain a large number of samples with missing views, while it can be difficult to obtain SAR images of all predefined views during testing. The way existing models handle complex and variable multiview samples is of great significance to their practicality. A commonly used approach is to adopt certain filling strategies to transform the incomplete multiview problem into a complete multiview problem. However, strategies such as filling with monochromatic images or noise images filling may affect the feature fusion of other perspectives. Based on the assumption of multiview image correlation, some popular methods use generative models to predict and complete missing view images or embeddings by using other views, but this increases the overall model complexity and does not inherently improve recognition performance. Our idea is to minimize the impact of missing views on feature fusion; therefore, we propose a multiview semantic prototype completion method. The feature clustering center of a large number of samples in a certain category is the prototype of the category, which has rich and stable semantic meanings. Based on the proposed SeVi model, semantic and view prototype vectors are calculated and dynamically updated during the training process. Then, the prototypes are used to replace the embeddings of missing views, enabling the training and inference of multiview recognition models under the condition of incomplete views.

Another practical challenge in multiview SAR target recognition is the scarcity of multiview recognition data. Capturing images from multiple views of the same target requires high demands on both the sensor platform and the target, making it difficult to collect a large number of target samples [19]. Due to potential background clutter interference or the lack of clear semantic features in SAR images of targets from certain views, it is difficult to determine the true view unless there is extensive expert experience. Since we are only using the view information to assist with the recognition task and are not concerned with the precise observed view of the image, we only need to label a small number of clear views. During training, the predicted distribution of views provided by the network can also serve as effective view embedding information. Based on this idea, we introduce a semi-supervised

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learning (SSL) based view classification method, which can achieve high performance on view classification with a small amount of view labels. Therefore, we construct a nonstrict multiview recognition dataset by using existing target recognition datasets. This approach makes full use of existing SAR ATR data resources, alleviating the problem of limited data for multiview SAR ATR tasks.

In summary, we propose a multiview target recognition framework, and our contributions are mainly threefold.

1) We construct a multiview feature embedding fusion network *SeVi* that achieves highly flexible and robust multiview image feature fusion and high accuracy of target recognition.
2) We propose a multiview semantic completion strategy that enables the proposed model to train and infer under incomplete view conditions while maintaining high performance.
3) We introduce a method for constructing nonstrict multiview datasets based on semi-supervised classification, which alleviates the issue of data scarcity in multiview SAR target recognition.

The remainder of the article is organized as follows. Section II introduces related work on multiview learning. Section III first presents the semantic-view dual-branch embedding fusion network model (*SeVi*), followed by the multiview semantic prototype completion method proposed to address the issue of incomplete view learning, and then discusses the semi-supervised view classification and nonstrict multiview dataset construction methods proposed to tackle the problem of multiview data scarcity. Section IV provides a detailed description of the experimental work and analysis. Section V concludes this article and provides an outlook for future work.

II. Related Work

A. Multiview Automatic Target Recognition

Multiview target recognition relies on the consistency and complementarity of target features across different views. For optical images, the features such as color, structure, and texture of a target under different views are highly correlated. Traditional multiview recognition methods for the optical modality mostly extract the common features of views through feature decomposition methods for recognition. For SAR images, however, changes in observation views will significantly reduce the correlation of target features. Moreover, due to differences in SAR imaging platforms, imaging modes, and imaging parameters, the SAR image features of the same target can be completely distinct. Therefore, the SAR modality mainly leverages the complementarity of features obtained under different observation configurations for recognition, and fusing multiple view features through deep learning models for recognition has become a mainstream research direction.

Early SAR multiview ATR models mainly adopted a fusion architecture of CNN and fully connected layers. Pei et al. [20] studied the joint recognition capability of monostatic SAR and bistatic SAR with separate transmitters and receivers, and improved the target recognition accuracy by fusing the features of monostatic and bistatic observed images through a simple CNN model. Study [7] conducted research on the optimization of the flight path of the SAR platform and multiple observation perspectives, and adopted CNN channel concatenation to realize the fusion of feature maps from two selected perspectives. With the advent of LSTM, LSTM-based fusion models have garnered significant attention for their capacity to capture interview relationships. Wang et al. [21], [22] considered obtaining sequential images of the target from different azimuths, and propose a hybrid CNN-LSTM architecture, using CNN to extract feature vectors and LSTM for multiview fusion. Pei et al. [23] used a recurrent neural network (RNN) for multiview fusion, transforming parallel multiview fusion into progressive fusion between new view features and existing features, solving the problem of fixed input angles. It introduces a gated network structure to adaptively learn the weights of new views and fused feature maps, thereby learning which views are more beneficial for target recognition. RNN structures, such as LSTM, are inherently designed for sequence modeling, requiring view images to be input in a sequential manner and exhibiting a forgetting mechanism. In image-related tasks, LSTM-based fusion architectures have gradually been replaced by Transformer. The Transformer structure, which can model global dependencies, has been increasingly applied to multiview and multimodal feature fusion processing [24], [25]. Li et al. [26] proposed a CNN-Transformer architecture, utilizing autoencoders to extract image feature vectors and a Transformer to fuse these vectors. The fusion of views mainly involves images with similar observation azimuths, which helps stabilize the features of sequential images. References [7], [20], [21], [22], [23], [26] mainly focus on the scenario of continuous circular observation of targets by airborne SAR platforms. Therefore, they still focus on the fusion of image sequences from adjacent views, which have high requirements on input conditions, and do not consider cross-view situations.

Multiview feature fusion recognition models are more complex and pose greater challenges in training compared to single-view recognition models. Researchers have adopted techniques such as constraint loss and data augmentation from few-shot learning to enhance the multiview feature learning capability. Under few-shot training conditions, Dai et al. [27] proposed a storable Capsule structure that reduces intraclass distances and increases interclass distances by combining features from multiple samples. Lui et al. [28] introduced a multiview semantic data augmentation method to complement sparse view data, enhancing the representation ability of the model. Xiao et al. [29] proposed a feature-based data augmentation method to improve recognition performance under sparse view conditions. In the first stage, features from images with different views are extracted, and the feature center is calculated. Then, feature augmentation is performed by combining the original image features with the feature center. In the second stage, the augmented features are used to train the classifier. Lv et al. [30] proposed a method for joint training of the same-class multiview and different-class samples, increasing the interclass differences. Zhang et al. [31] employed dimensionality reduction and sparse representation methods, designing view loss, semantic loss, and coupling

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loss functions to jointly optimize both intraclass and interclass distances. Zhang et al. [18] introduced a multiview feature fusion method based on CNN channel concatenation, using an attention mechanism to calculate weighted values.

Beyond classification tasks, another important application scenario of multiview fusion is the Reidentification task. Due to the significant differences in target images under different views, view prior information is usually introduced in model design, which provides valuable insights. Teng et al. [17] divided viewpoints into three categories and constructed a three-branch feature extraction module, along with a viewpoint classification module to predict classification probabilities and perform decision fusion on the extracted features. Teng et al. [32] classified viewpoints into four categories and introduced viewpoint adversarial training loss to reduce intraclass variations caused by different viewpoints. Sun et al. [33] utilized prior knowledge to detect key vehicle components and perform local feature extraction in component regions, addressing the problem of significant changes in overall image differences caused by viewpoints. Wang et al. [34] divided viewpoints into eight categories and fused viewpoint features with appearance features using an attention mechanism, effectively reducing issues caused by camera viewpoint differences.

We can conclude that for multiview SAR ATR tasks, the Transformer structure is currently the mainstream for feature fusion architectures. This is because it possesses the ability to model global information, enabling it to better learn the relationships between different views. In addition, multiview recognition models can further enhance their capability to model multiview features through methods such as loss function design and the introduction of view prior information.

B. Incomplete Multiview Learning

In practical multiview recognition tasks, complex incomplete multiview scenarios are often encountered. This is because during data collection and actual testing, factors, such as limitations of the observation system, errors in the processing flow, and noncooperative targets, may lead to failure in acquiring certain views or severe interference to view images. One of the underlying assumptions of most existing multiview recognition models is that all views of the target can be observed or a sufficient number of views need to be guaranteed, thus limiting the practicality of the models. Researchers have paid attention to multiview learning under incomplete view conditions. Traditional methods primarily include matrix factorization, kernel learning, etc., with the main idea being to obtain consistent representations across multiple views and use the common features from multiple views to recover incomplete views or perform other downstream tasks [35]. The study in [36] assumes that different views are generated by a shared subspace and uses matrix completion methods to recover the missing views, proving the convergence of the solution method. Methods based on deep learning mainly use generative methods, such as autoencoders and generative adversarial networks, to recover incomplete views. The study in [37] proposes that a complete latent representation in multiview learning can recover information from all perspectives and uses adversarial learning strategies to enhance the latent space semantic representation and reconstruction ability for missing views. In addition, some studies directly target downstream tasks like prediction or classification, using existing views to fuse information without recovering missing views, or employ certain imputation strategies to reduce the impact of missing views on the other complete views. The study in [38] constructs specific encoder–decoder architectures for each view and designs a weighted fusion module to minimize the negative effects of incomplete cases. Xie et al. [39] argues that missing views are highly uncertain, and providing a single imputation is unreasonable. Therefore, it proposes using multiple samples from a Gaussian distribution to impute the missing views and then applies an evidence-based fusion strategy to select the imputed view with the highest confidence.

The imputation strategy for missing views has a significant impact on multiview learning. For classification tasks, if simple solid-color images are used to fill in missing views, it is equivalent to introducing an image representing a special semantic category, which clearly disrupts the original fusion of multiview features [35]. We believe that using the mean vector of various semantic prototypes, i.e., the semantic clustering centers, as the high-dimensional embedding for the missing views can minimize the impact of missing views on the fusion of other views. Clustering center methods are commonly used in few-shot learning [40] and unsupervised representation learning [41]. Prototypical Networks are based on the idea that there is an embedding scheme where the points of each category cluster around the prototype representation of that category. By mapping the samples into the latent space, classification can be achieved by calculating the distance between the query point and the category prototype [40]. SWaV uses clustering centers to replace numerous unstable negative samples for contrastive learning in unsupervised visual representation learning, which can effectively enhance the unsupervised feature learning ability, showing that clustering centers have rich and stable semantic meanings [41]. In [42], for the task of view matching, a prototype imputation strategy is proposed to address the issue of missing instances, which can alleviate the instability of simple imputation under high missing rates.

For incomplete multiview SAR ATR tasks, generative model-based methods utilize other views to generate the missing views' images. However, constructing and training such models are relatively complex, and they may introduce biases that affect the recognition task. In contrast, complementing missing views using clustering centers of view embeddings is more stable and does not rely on additional models or training steps. Therefore, using prototypes to complement missing views is an effective approach to enhance the practicality of multiview recognition models.

C. Semi-Supervised Classification

To train a multiview recognition model on existing SAR target recognition datasets with abundant samples, we first manually label a small amount of multiview data based on factors, such as target structure and image features. Then, we efficiently complete the view label assignment using

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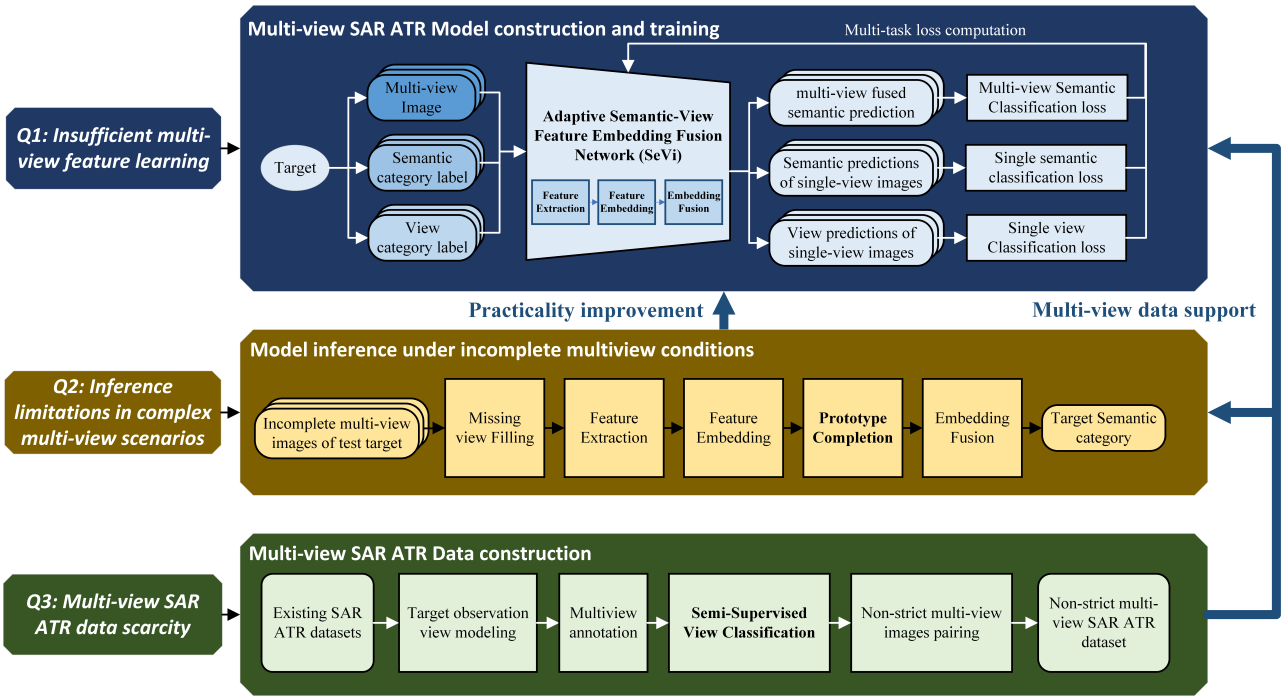

Fig. 1. Framework of the multiview SAR ATR method based on adaptive semantic-view embedding fusion network.

semi-supervised classification methods. Finally, we combine the multiview images to create a nonstrict multiview target recognition dataset. Since view classification tasks have strong human experience bias, unsupervised clustering methods [43] are challenging to rely solely on image semantics to cluster data by view. In the SSL framework, the model is first trained on the labeled data and then used to predict pseudo-labels (PLs) for the unlabeled data. The model is then trained based on both the true labels and pseudo-labels to achieve optimal performance across all data. SSL methods can be summarized as enhancing the similarity of latent space points [44]. Deep learning-based SSL methods can be divided into three main types: 1) pseudo-labeling; 2) teacher-student models; and 3) consistency regularization.

The pseudo-labeling method transforms high-confidence predicted categories of unlabeled data into pseudo one-hot labels and then trains the model in a supervised manner on both labeled and unlabeled data. The teacher-student model is an iterative update method involving two models: first, the teacher model is trained on labeled data, then the teacher model is used to infer pseudo-labels for unlabeled images; the student model is trained on the combined data, and after training, the student model serves as the new teacher model. The basic idea of consistency regularization is that different versions of the semantic representation of an input image, generated through data augmentation or fine-tuned models, should remain the same. Specifically, data augmentation plays an important role in representation learning and can greatly improve model performance [45]. The paper [46] proposed the FixMatch network, which achieves superior performance by using both pseudo-labels and consistency loss modules in the baseline. Further optimization of pseudo-label selection and consistency regularization based on FixMatch can lead to even higher performance [47], [48], [49].

We can conclude that the key components of semi-supervised classification methods include pseudo-labels, consistency regularization, and data augmentation techniques, which together enable the completion of classification tasks under the combined effect of a small amount of label supervision and representation learning. However, existing studies are mainly based on optical images, and the data augmentation methods are designed for three-channel RGB data. Therefore, we plan to adopt the FixMatch model and preliminarily explore data augmentation methods suitable for SAR images to efficiently complete the view annotation of SAR target images, thereby supporting the training of multiview recognition models. Therefore, data augmentation methods and models suitable for semi-supervised classification of SAR images still need further exploration to achieve efficient annotation of SAR target image views.

III. Methodology

In this section, we present the proposed multiview SAR ATR method. As illustrated in Fig. 1, we propose an adaptive semantic view embedding fusion network, along with a multi-view SAR ATR framework to address three key challenges in multiview SAR ATR tasks: 1) insufficient multiview feature learning; 2) inference limitations in incomplete multiview scenarios; and 3) scarcity of multiview SAR ATR data.

First, to tackle the issue of insufficient multiview feature learning for targets, we construct a feature-level fusion network, which incorporates view prior information via view labels to assist the model in learning multiview features, and adopts multitask learning for end-to-end training and optimization. In Section III-A, the multiview recognition problem is defined; based on the definition, Section III-B provides a detailed description of each component of the proposed model.

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Second, it is difficult to acquire ideal view images of the same target in practical applications. Consequently, multiview feature-level fusion recognition models need to be applicable under view-missing conditions. To address the model inference challenge in complex multiview scenarios, we propose a prototype-based view completion method. This method uses prototypes to replace missing view embeddings for fusion, ensuring the model remains applicable and maintains high performance in complex scenarios. Section III-C elaborates on the missing-view prototype completion algorithm based on the SeVi model.

Third, multiview SAR ATR faces challenges in data acquisition and annotation. To address the data scarcity issue, we propose a method for constructing a nonstrict multiview dataset. Specifically, we perform discrete modeling on the target observation views, conduct view annotation on existing SAR ATR datasets, and use SSL to efficiently complete view classification. Subsequently, we obtain nonstrict multiview SAR ATR datasets through multiview pairing to support model training. Section III-D details the semi-supervised SAR view classification method.

A. Multiview Recognition Problem Formulation

First, we build the geometric model of the multiview SAR ATR problem. In general, sensors for multiview remote sensing observations can be multimodal (e.g., optical images or SAR images) and from heterogeneous platforms (e.g., airborne and spaceborne platforms). In this article, we primarily focus on observations from multiple azimuthal angles using the SAR modality from a single platform or several same-type platforms, avoiding significant domain biases between images from different views due to differences in resolution or noise, which is not the focus of this article. Typically, airborne or spaceborne SAR platforms achieve 2-D focused imaging of targets in range and azimuth directions through platform motion. A typical geometric model for multiview SAR observations is shown in Fig. 2(a). When the radar antenna illuminates a target from different azimuth angles, the backscattering coefficients of the target vary, leading to significant differences in the scattering structure features observed in the target images. We categorize azimuth angles into four views: forward view, forward squint view, side view, and backward squint view, as illustrated in Fig. 2(b). Note that to ensure each view category covers an equal angular range, the forward and backward views are grouped together; therefore each view spans 90°. This categorization is motivated by two key considerations: First, most vehicle or aircraft targets exhibit axial symmetry. Second, SAR image features of targets are determined not only by the signals reflected from their own scattering structures but also by the interaction between the target and the background. The dihedral angles formed between the front, back, or sides of the target and the ground surface produce distinct scattering characteristics when illuminated by radar waves at vertical or squint angles. In addition, shadows caused by target occlusion also vary with the observation view. We model views using discrete rather than continuous functions for several reasons: First, a few well-chosen views with significant angular differences are

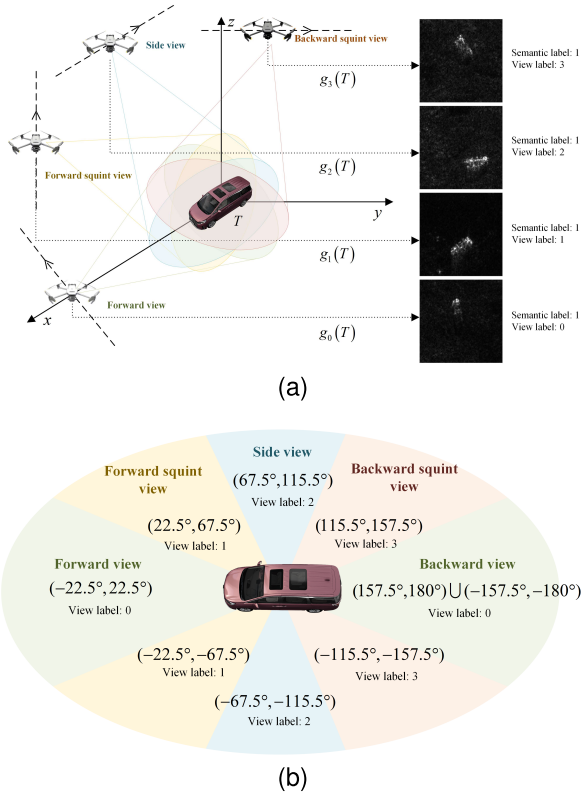

Fig. 2. Typical multiview SAR observation geometric model. (a) Four-category classification of azimuth angles and corresponding SAR image examples. Multiview SAR images of Buick GL8 in ATRNet-STAR dataset [6]. (b) Classification of azimuth viewing angle categories.

usually sufficient to capture target features adequately. Second, acquiring fine-grained azimuth-angle images for SAR target observation is time-consuming and resource-intensive, and adopting a discrete model serves as a compromise solution. Furthermore, our primary use of view features is to identify semantic relationships between images from different views to guide semantic fusion across views, which does not require high-precision view regression, which will be discussed in detail later. Therefore, we model observation views by comprehensively considering both recognition performance and implementation feasibility.

Next, we introduce the multiview recognition problem formulation. We define a target as T, whose semantic category is defined as $y \in S_s : \{0, 1, \ldots, N_s - 1\}$, where S represents the set of semantic categories and N_s represents the number of semantic categories. The multiview imaging process can be modeled as a function $g(T, v)$ with the target T and the view v as parameters. For multiview imaging involving N_v views, it can be expressed as

$$\{I_i\}_{i=0}^{N_v-1} = \{g(T, v = i)\}_{i=0}^{N_v-1}. \tag{1}$$

Multiview imaging can be regarded as a process of extracting multiple features of the target. Different from the natural background, the artificial targets to be recognized usually have regular geometric shapes, sizes, orientations, textures, shadows, etc. The features exhibited by the targets vary significantly under different observation views. Ideally, we explicitly know which target attributes are consistent or complementary

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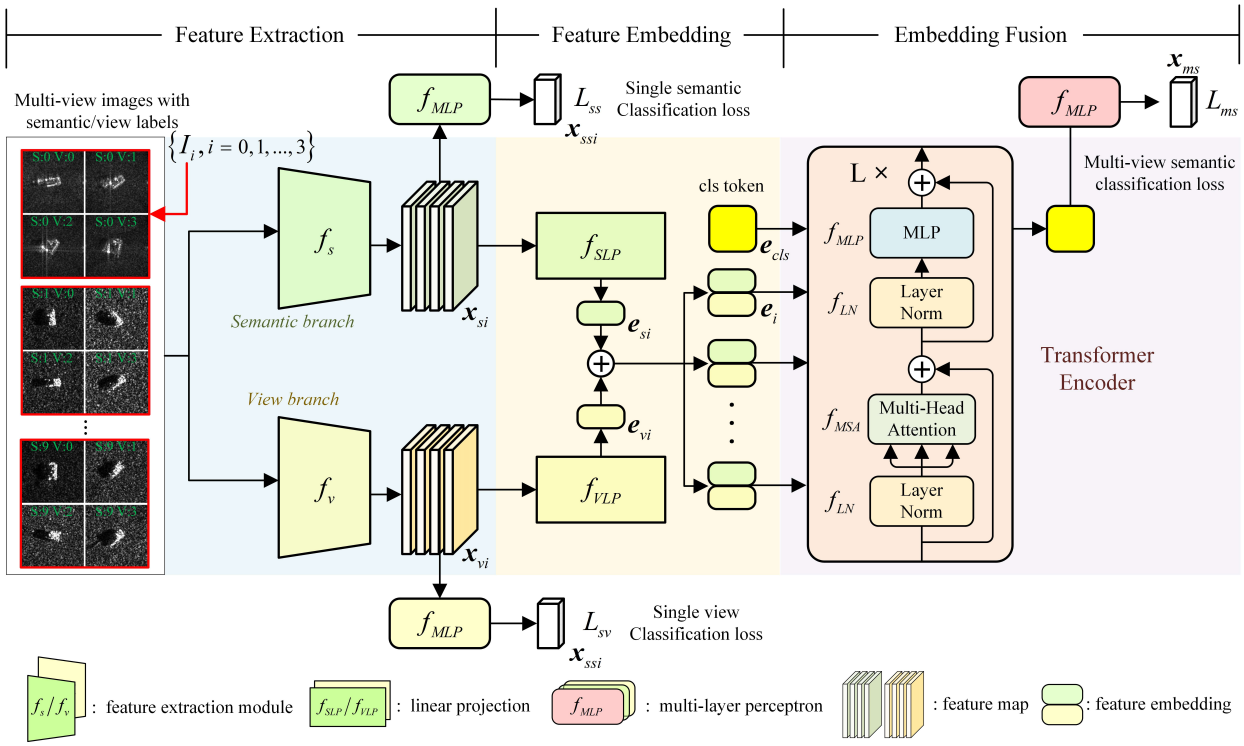

Fig. 3. Framework of the *SeVi* model. An example of the model input is provided on the left. The red box represents a target, which contains images from four views; each image has a corresponding semantic label S and view label V; multiple targets are grouped into a batch during training. The model consists of three stages. In the feature extraction stage, the semantic feature maps and view feature maps of the images from each view are obtained, respectively. Auxiliary classification heads are constructed to guide the feature extraction during training. In the feature embedding stage, the view-aware embeddings of the images from each view are obtained. In the embedding fusion stage, the self-attention mechanism of the Transformer is used to achieve the information interaction among different views. Finally, the category token is used for recognition.

among different views. However, this usually depends on complex semantic labels. But it is relatively easier for us to confirm the viewing angles of the observed targets according to the relative relationship between the sensor and the target, and assign one-hot labels to the images of different views to assist the multiview recognition task. As shown in Fig. 2, according to the observation angle or the position of the sensor, a view label can be assigned to a single-view image as $v \in S_v : \{0, 1, \ldots, N_v - 1\}$, where S_v represents the set of view categories and N_v represents the number of views.

The multiview image recognition task is to train a classifier f_θ with parameters θ based on distinct multiview images to predict the semantic category of the target, which can be expressed as

$$\hat{y} = f_\theta \left(I_0, I_1, \ldots, I_{N_v - 1} \right) \tag{2}$$

where $\hat{y}$ denotes the predicted semantic label of the target.

B. Adaptive Multiview Embedding Fusion Network

Here, we introduce the core network model proposed in this article. The architecture diagram of the semantic-view dual-encoder feature embedding fusion network termed *SeVi* is shown in Fig. 3. *SeVi* has three main designs: 1) the dual-branch feature extraction and embedding module; 2) the multiview embedding fusion module; and 3) the multitask learning loss function.

In the training phase, the input of *SeVi* is a set of targets. Each target contains N_v images from different views, and each image has both a semantic label and a view label. First, each single-view image of the target passes through the dual-branch feature extraction module to obtain the semantic feature map and the view feature map. Then, the semantic feature map and the view feature map respectively pass through the linear projection module to complete the feature embedding. The semantic embedding and the view embedding are concatenated in a channel, resulting in the feature embedding of the single-view image. Note that the parameters of each branch are independent. The number of feature embeddings is equal to the number of images with different views.

The multiview embedding fusion module adopts a structure similar to that of ViT [50]. A class token embedding is constructed and concatenated with a multiview feature embedding. Since the prior view information has been used to perform view embedding on the images, which represents the relative relationships among the multiview images, there is no need for additional position embedding. Then, the multiview feature embedding is then input into the Transformer encoder blocks, and multiview feature fusion is accomplished by the multihead self-attention mechanism. The output class token embedding serves as the multiview feature vector.

To enable end-to-end training, *SeVi* employs three types of hybrid loss functions. The semantic feature map, view feature map, and the multiview semantic feature vector are passed through the corresponding MLP classification heads, after which the single-view semantic classification loss, single-view classification loss, and multiview semantic classification

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loss based on cross-entropy are calculated. By fine-tuning the weights of the hybrid loss functions, the training stability and model performance can be enhanced.

1) Semantic-View Dual-Branch Feature Extraction: For a typical multiview target recognition task, the single-view image is denoted as $I_i \in \mathbb{R}^{C\times H\times W}, i = 0, 1, \ldots, N_v - 1$, where N_v denotes the number of views and C, H, and W denote the numbers of channels, height, and width of the original image, respectively. The dual-branch feature extraction module separately extracts semantic and view features from the input multiview images. The module structure consists of two identical CNNs with independent parameters. We adopt ResNet50 [51] as the backbone and the baseline model. Then, we can obtain

$$\begin{aligned} \boldsymbol{x}_{\mathrm{si}} &= f_s\left(I_i;\theta_s\right) \\ \boldsymbol{x}_{\mathrm{vi}} &= f_v\left(I_i;\theta_v\right) \end{aligned} \tag{3}$$

where f_s denotes the semantic feature extraction module with learnable parameters θ_s; f_v denotes the view feature extraction module with learnable parameters θ_v. $\boldsymbol{x}_{\mathrm{si}}, \boldsymbol{x}_{\mathrm{vi}} \in \mathbb{R}^{C_f\times H_f\times W_f}$ denote the semantic feature map and the view feature map of the single image I_i, where C_f, H_f, and W_f denote the number of channel, height, and width of the output feature maps.

2) Multiview Feature Embedding Fusion: We design a multiview embedding fusion module based on the ViT network and utilize the multihead self-attention mechanism of the Transformer encoder to achieve the fusion of feature embeddings of different view images. First, we use a linear projection module composed of a two-layer convolutional network to perform channel compression to reduce the number of parameters and perform embedding to meet the input requirements of the Transformer encoder, which can be expressed as $f_{\mathrm{LP}}(\cdot) = f_{\mathrm{conv2}}(f_{\mathrm{conv1}}(\cdot))$, where $f_{\mathrm{conv}}(\cdot)$ denotes a convolutional layer. The first layer is a 1×1 point-wise convolutional layer, and the output channels of the first layer generally take a smaller value to reduce the number of network parameters. The size of the convolutional kernel of the second layer is the same as that of the feature map, which realizes transforming the 3-D feature map into a 1-D embedding. Then, we can obtain

$$\begin{aligned} \boldsymbol{e}_{\mathrm{si}} &= f_{\mathrm{SLP}}\left(\boldsymbol{x}_{\mathrm{si}}\right) \\ \boldsymbol{e}_{\mathrm{vi}} &= f_{\mathrm{VLP}}\left(\boldsymbol{x}_{\mathrm{vi}}\right) \end{aligned} \tag{4}$$

where $\boldsymbol{e}_{\mathrm{si}}, \boldsymbol{e}_{\mathrm{vi}} \in \mathbb{R}^{D/2}$ represent the semantic embedding and the view embedding of the input single-view image $\boldsymbol{I}_i$. Then, we perform concatenation in the dimension of vector length on the embeddings to obtain the feature embedding of the corresponding single-view image

$$\boldsymbol{e}_i = \boldsymbol{e}_{\mathrm{si}} \oplus \boldsymbol{e}_{\mathrm{vi}} \tag{5}$$

where $\boldsymbol{e}_i \in \mathbb{R}^D$; $\oplus$ denotes channel concatenation. Then, the learnable class token is constructed and stacked together with the feature embeddings of the multiview images to obtain the embedding input of the Transformer encoder, which can be expressed as

$$\boldsymbol{e} = \left[\boldsymbol{e}_{\mathrm{cls}}; \boldsymbol{e}_1; \boldsymbol{e}_2; \cdots; \boldsymbol{e}_{N_v}\right] \tag{6}$$

where $\boldsymbol{e} \in \mathbb{R}^{N_e\times D}$, $N_e = N_v + 1$.

The Transformer encoder is stacked by L layers of blocks, and the main expression of layer $l \in (0, 1, \ldots, L-1)$ is

$$\begin{aligned} \boldsymbol{e}' &= f_{\mathrm{MSA}}\left(f_{\mathrm{LN}}\left(\boldsymbol{e}^{(l-1)}\right)\right) + \boldsymbol{e}^{(l-1)} \\ \boldsymbol{e}^{(l)} &= f_{\mathrm{MLP}}\left(f_{\mathrm{LN}}\left(\boldsymbol{e}'\right)\right) + \boldsymbol{e}' \end{aligned} \tag{7}$$

where $f_{\mathrm{LN}}(\cdot)$ denotes layer normalization; $f_{\mathrm{MSA}}(\cdot)$ denotes multihead self-attention; f_{MLP} denotes multilayer perceptron. The embeddings of multiview images are fused through the global multihead self-attention mechanism, which can be expressed as

$$\mathrm{Head}_i = \mathrm{softmax}\left(\frac{\boldsymbol{q}_i \boldsymbol{k}_i^T}{\sqrt{D_h}}\right)\boldsymbol{v}_i, i = 0, 2, \ldots, h-1 \tag{8}$$

where $\boldsymbol{q}_i, \boldsymbol{k}_i, \boldsymbol{v}_i \in \mathbb{R}^{N_e\times D_h}$ are the query, key, and value vector of the ith head obtained by passing the embedding $\boldsymbol{e}$ through a linear layer, copying it into three parts, and then evenly dividing it into multiple heads. h is the number of heads. $D_h = D/h$ is the feature dimension of each head. After concatenating the outputs of all heads, they are mapped to the original dimension through a linear layer. The output size of each Transformer encoder block is $N_e \times D$. The output of the last Transformer encoder block is further processed by layer normalization, and the feature vector corresponding to the class token in the output is taken as the output of the multiview feature fusion module, which can be expressed as $\boldsymbol{e}_{\mathrm{cls}}^{(L-1)} \in \mathbb{R}^D$.

3) Hybrid Semantic-View Loss Function Design: The hybrid loss function is composed of three parts, specifically including: the sum of single-view semantic classification losses, view contrastive loss, and multiview semantic classification loss. Based on the cross-entropy loss, the single-view semantic classification loss can be expressed as

$$\mathcal{L}_s = -\sum_{i=0}^{N_v-1}\sum_{j=0}^{N_s-1} y_{\mathrm{sj}} \log p\left(\hat{y}_{\mathrm{sj}}|\boldsymbol{z}_{\mathrm{si}}\right) \tag{9}$$

where y_{sj} is the jth semantic class label, $p(\hat{y}_{\mathrm{sj}}|\boldsymbol{z}_{\mathrm{si}})$ is the prediction probability of the jth semantic class, and $\boldsymbol{z}_{\mathrm{si}}$ represents the semantic feature vector output of the ith view obtained by the single-view semantic classification head. The view contrastive loss can be expressed as

$$\begin{aligned} \mathcal{L}_v = &-y_{\mathrm{vk}_0} \log p\left(\hat{y}_{\mathrm{vk}_0}|\boldsymbol{z}_{\mathrm{vk}_0}\right) \\ &+ \sum_{k=0,k\neq k_0}^{N_v-1} \max\left(\mathrm{margin} - \left\|\boldsymbol{z}_{\mathrm{vk}} - \boldsymbol{z}_{\mathrm{vk}_0}\right\|^2, 0\right) \end{aligned} \tag{10}$$

where y_{vk} denotes the kth view class label, $p(\hat{y}_{\mathrm{vk}}|\boldsymbol{z}_{\mathrm{vk}})$ denotes the kth prediction probability of the kth view. k_0 represents a randomly selected view. $\boldsymbol{z}_{\mathrm{vk}}$ denotes the feature vector output of the kth view obtained by the view classification head. This loss enables the semantic classification task and the view classification task to be carried out simultaneously. Since each target input contains various views, if the sum of cross-entropy losses from the classification of each view is adopted, the view branches cannot quickly capture the differences between views, resulting in low efficiency and instability in model training. Therefore, contrastive learning is introduced. For the input of each target, a view is randomly selected, the cross-entropy loss of classification for this view is calculated, and at

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the same time, the contrastive loss between this view and other views is computed. This approach improves the efficiency of model training and the capability of view classification.

Moreover, the multiview fusion semantic classification loss can be expressed as

$$\mathcal{L}_m = -\sum_{j=0}^{N_s-1} y_{sj} \log p\left(\hat{y}_{sj}|\boldsymbol{z}_m\right) \tag{11}$$

where $\boldsymbol{z}_m$ denotes the semantic feature vector obtained by the multiview semantic classification head. Then, we can obtain the total loss as

$$\mathcal{L} = \alpha_s \mathcal{L}_s + \alpha_v \mathcal{L}_v + \alpha_m \mathcal{L}_m \tag{12}$$

where α_s, α_v, and α_m are hyperparameters. We tested several groups of values with different proportional coefficients in experiments and found that these three parameters are not sensitive to the results within a certain proportional range. Therefore, without loss of generality, the proportion of the loss function coefficients is set to 1: 1: 1.

C. Prototype Learning for Incomplete Multiview Recognition

Incomplete multiview completion is of twofold significance for multiview recognition models. First, it meets the input requirements of the network model and improves the stability of model inference. Second, it enables effective utilization of incomplete multiview data. In practical ATR tasks, it is usually difficult to obtain all the preset multiview samples of each target, which means that there are a large number of incomplete multiview samples in the dataset. If the model cannot properly handle the situation of missing views or effectively utilize the large amount of incomplete multiview sample data, the model's performance in practical tasks will be significantly degraded.

Commonly used incomplete view completion methods mainly include monochrome image filling, noise image filling, and filling with other views, which can meet the input requirements of multiview recognition models. Monochrome or noise image filling essentially introduces noise features, which have neither spatial structure nor semantic information, and thus exhibit a large distribution difference from the features of actual view images. Filling with other views essentially introduces features with view semantic misalignment, which amplifies the inherent distribution bias of the existing views. In Transformer-based fusion models, the bias introduced by the tokens of the completed missing views will spread to the entire model through the calculation of the cross-attention mechanism, making the result of fusion classification uncontrollable. Therefore, it is of great importance to reduce the bias of the filled features for missing views.

To address the issue of incomplete multiview inference, we propose a multiview prototype completion method. Our intuitive idea is to use the semantic and view clustering center vectors to replace the embeddings of the missing views. The prototype can retain shared semantic information and possesses more stable and controllable semantic features. The prototype completion approach can reduce the errors introduced by the missing views and the impact on the multiview embedding fusion from other views, thereby enhancing the robustness and accuracy of the fusion results.

The computation of semantic prototypes and view prototypes is crucial for the prototype completion method. Theoretically, prototypes should be obtained by averaging the feature vectors of a sufficient number of class-balanced samples to minimize bias as much as possible. However, during the training of the multiview recognition model, the number of samples in a single batch is too small, and there exists a class imbalance issue. As a result, prototypes computed on a single batch of data have large biases, and with the update of network parameters, the prototypes computed for each batch may be unstable. Moreover, computing prototypes for all samples during each batch training is too computationally expensive. Therefore, to address the above issues, we construct a set of sufficiently large and dynamically updated circular queues for computing semantic and view prototypes. Storing data from multiple batches through circular queues reduces the bias in prototype computation and improves stability.

The schematic of the incomplete multiview completion method is shown as Fig. 4. To meet the input requirements of the multiview feature embedding fusion network, we generate a view mask for each image. Practically, it is a one-hot encoded label, where the missing view is marked as 0, and a monochrome image is used as input instead; the existing view is marked as 1. During the training process, one batch of data is input and passes through the semantic branch and view branch, respectively, to obtain the semantic embeddings and view embeddings of the images from existing views.

The clustering center of the semantic embeddings of the current batch can be calculated as

$$\bar{\mathbf{e}}_s^{(i)} = \frac{1}{\sum_{j=0}^{B-1} \mathbb{1}_{y_j==i}} \sum_{j=0}^{B-1} \mathbb{1}_{y_j==i} \frac{\sum_{k=0}^{N_v-1} m_{(i,k)} \mathbf{e}_{s(i,k)}}{\sum_{k=0}^{N_v-1} m_{(i,k)}} \tag{13}$$

where B represents the batch size, $\mathbb{1}_{y_j==i}$ takes the value 1 when the label of the jth sample y_j is the semantic category i, and 0 otherwise. $m_{(j,k)}$ and $\mathbf{e}_{s(j,k)}$ represent the view mask and the semantic embedding of the kth view of the jth target data in this batch.

For each semantic category, we construct a circular queue $\mathbf{Q}_s^{(i)} \in \mathbb{R}^{N\times D}$, $i = 0, 1, \ldots, N_s - 1$, where N is the length of the queue, and D is the length of the embedding vector. Usually, we set a relative large value for N to ensure that the queue contains a sufficient number of samples for prototype calculation and avoid domain knowledge bias. During training, the clustering centers of semantic embeddings in the current batch are added to the queue, which can be expressed as $\mathbf{q}_{s(n)}^{(i)} \leftarrow \bar{\mathbf{e}}_s^{(i)}$. When the queue is full, as the latest clustering center of semantic embeddings enters the queue, the earliest clustering center of semantic embeddings will be removed from the queue, thus achieving dynamic updates. By calculating the mean of the vectors in the circular queue, the semantic prototype of a certain category can be obtained as

$$\mathbf{p}_s^{(i)} = \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{q}_{s(n)}^{(i)} \tag{14}$$

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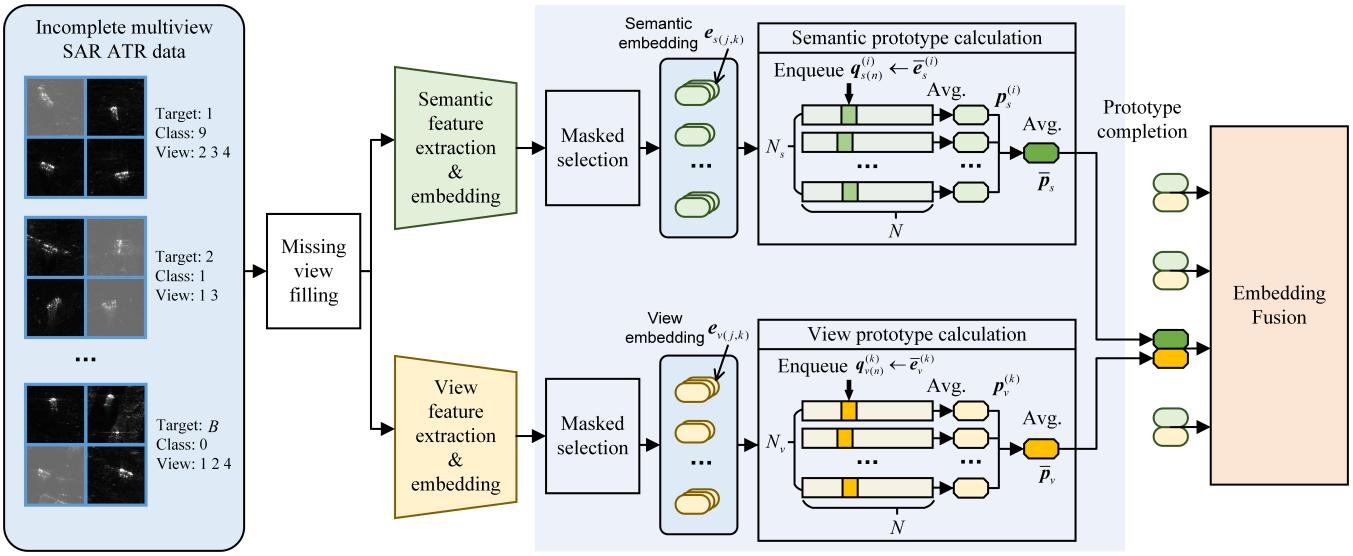

Fig. 4. Framework of the incomplete multiview filling strategy. The input data of the model are multiview images with masks. First, a simple filling method is used to meet the input requirements of the model, and the embeddings are obtained through forward calculation. Then, the embeddings are selected according to the masks, and the clustering centers of each semantic category and view category in the current batch are calculated and added to the circular queue. Furthermore, the mean value of the circular queue is calculated to obtain the prototype for filling, and the filled embeddings of the images from each view are input into the subsequent fusion module.

where $\mathbf{q}_{s(n)}^{i}$ denotes the nth vector in $\mathbf{Q}_s^{(i)}$. Furthermore, by computing the mean value of the semantic prototypes of each semantic category, the average semantic embedding can be obtained as

$$\overline{\mathbf{p}}_s = \frac{1}{N_s} \sum_{i=0}^{N_s-1} \mathbf{p}_s^{(i)}. \tag{15}$$

Thus, we can use $\overline{\mathbf{p}}_s$ to replace the semantic embedding of the missing view.

Similarly, for view embeddings, we construct circular queues for each view as $\mathbf{Q}_v^{(k)}$, $k = 1, 2, \ldots, N_v$. During training, first calculate the mean of the view embeddings of each view in the current batch, which is expressed as

$$\overline{\mathbf{e}}_v^{(k)} = \frac{\sum_{j=0}^{B-1} m_{(j,k)} \mathbf{e}_{v(j,k)}}{\sum_{j=0}^{B-1} m_{(j,k)}} \tag{16}$$

where $\mathbf{e}_{v(j,k)}$ represents the kth view embedding of the jth target in this batch. Then, the clustering centers of view embeddings for each view in the current batch are added to the circular queue, which can be expressed as $\mathbf{q}_{v(n)}^{(k)} \leftarrow \overline{\mathbf{e}}_v^{(k)}$. Next, calculate the mean of all vectors in each queue to obtain the mean of the view embeddings corresponding to a certain view, which is used to fill in the view embeddings of the missing views. The prototype of the view embeddings of the kth view in the batch is

$$\mathbf{p}_v^{(k)} = \frac{1}{N} \sum_{i=0}^{N-1} \mathbf{q}_{v(n)}^{(k)} \tag{17}$$

where $\mathbf{q}_{v(n)}^{(k)}$ represents the nth vector in the circular queue $\mathbf{Q}_v^{(k)}$. Furthermore, the average view prototype for filling can be obtained as

$$\overline{\mathbf{p}}_v = \frac{1}{N_v} \sum_{k=0}^{N_v-1} \mathbf{p}_v^{(k)}. \tag{18}$$

Based on the above derivations, we can obtain the key training process for incomplete multiview recognition as shown in Algorithm 1. The proposed method decouples the calculation of the prototype vectors during training from the calculation of each batch. Although multiple mean value calculations increase the computational cost to a certain extent, they reduce the requirements for additional storage space and can overcome the problem of class imbalance.

Then, from the perspective of the error propagation mechanism, we conduct a comparative analysis of prototype completion and other completion methods. During model forward propagation, tokens from noise or monochrome image filling are irrelevant to real images. They may be incorrectly weighted in attention calculation, with random error perturbations contaminating the global representation. Tokens filled with images from other views contain target semantic information, but there may be significant discrepancies between repeated features and real features. This also tends to make attention overfit to a specific view, causing systematic biases. The essence of prototype completion for missing views is replacing individual-specific features with group-shared features. The error source is not semantic misalignment or noise contamination, but the loss of detailed information. Prototype completion is effective on the premise that data samples are unbiased and complete. In this case, semantic similarity can be measured by distance in the obtained feature space. Tokens from prototype completion have low error variance and still contribute to the completion of global features. Compared with methods such as noise image filling or other-view image filling, prototype filling provides a more stable and controllable feature distribution.

When there are missing data in the training set, during model backward propagation, the gradients obtained by the noise or monochrome image filling method are irregular,

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Algorithm 1 Key Training Process of Incomplete Multiview Recognition

Input: A batch of multiview images $\mathbf{I}$, semantic labels $\mathbf{y}$, view labels $\mathbf{v}$, view masks $\mathbf{m}$; number of views N_v; batch size B; size of circular queue N

Output: trained semantic embedding module f_s, view embedding module f_v, embedding fusion module f_e; average semantic prototype $\overline{\mathbf{p}}_s$, average view prototype $\overline{\mathbf{p}}_v$

Construct the semantic circular queue $\mathbf{Q}_s^{(i)}$, $i = 0, \ldots, N_s - 1$ and the view circular queue $\mathbf{Q}_v^{(k)}$, $k = 0, \ldots, N_v - 1$
for each batch **do**
  Replace the missing view with monochrome image:
  if $m_{(j,k)} == 0$ **then**
    $I_{(j,k)} = I_0,\ j = 0, \ldots, B-1,\ k = 0, \ldots, N_v - 1$
  end if
  Compute the semantic embedding $\mathbf{e}_{s(j,k)} = f_s(I_{(j,k)})$ and the view embedding $\mathbf{e}_{v(j,k)} = f_v(I_{(j,k)})$
  Compute the clustering centers of the semantic embeddings $\{\overline{\mathbf{e}}_s^{(i)}\}$, $i = 0, \ldots, N_s - 1$ and the clustering centers of the view embeddings $\{\overline{\mathbf{e}}_v^{(k)}\}$, $k = 0, \ldots, N_v - 1$
  Enqueue each of the clustering centers of the embeddings:
  for each circular queue $\mathbf{Q}$ **do**
    enqueue$(\overline{\mathbf{e}}, p)$ where p is the pointer of $\mathbf{Q}$
    $p = mod(p + 1, N)$
  end for
  Compute the semantic prototypes $\{\mathbf{p}_s^{(i)}\}, i = 0, \ldots, N_s - 1$ and the view prototypes $\{\mathbf{p}_v^{(k)}\}, k = 0, \ldots, N_v - 1$
  Compute the average semantic prototype $\overline{\mathbf{p}}_s$ and the average view prototype $\overline{\mathbf{p}}_v$
  Replace the embeddings of the missing view with the average prototypes:
  if $m_{j,k} == 0$ **then**
    $\mathbf{e}_{s(j,k)} = \overline{\mathbf{p}}_s,\ \mathbf{e}_{v(j,k)} = \overline{\mathbf{p}}_v$
  end if
  Continue with complete multiview fusion processing...
end for

making it hard to converge to an effective compensation strategy. For filling with other views, the corresponding gradients drive the model to over-rely on existing view features, further weakening its ability to model missing views. In contrast, for prototype filling, the gradients act directly on the embeddings generated by the prototype, pushing them to approximate the feature distribution of the actual missing view. As training proceeds, cross-view consistent prototypes can be formed, which enhances the robustness under incomplete inputs.

D. Multiview Data Construction Based on Semi-Supervised Classification

Another important practical problem faced by multiview recognition is that the matched multiview image data could be scarce. With the improvement of remote sensing observation and data processing capabilities, there is an increasing amount of ordinary target samples, covering multiple platforms, multitemporal, and multiview, etc. However, collecting strictly matched multiview images places higher demands on the capabilities of the sensor platform and the degree of target cooperation, making it difficult to obtain a large amount of sample data. Moreover, due to the special imaging mechanism of SAR, the multiview annotation of SAR targets highly depends on the empirical interpretation of professionals.

To address the above problems, we propose a method of using SSL for view classification of targets in the same semantic category. Our idea is mainly based on two aspects. First, we can make full use of existing data resources for SAR ATR to construct a nonstrictly multiview target recognition dataset to support training of multiview recognition models. Second, semi-supervised classification is the most feasible approach for the view classification task. This is because view classification is different from general semantic classification tasks. It is difficult to complete the view division from the semantic features of images through unsupervised methods such as deep clustering. In contrast, SSL can achieve high-performance classification by using a small number of labels.

First, we briefly introduce the definition of the semi-supervised classification problem. The ratio of the number of labeled samples to the total number of samples is defined as Γ. For a training batch with the batch size of B, the ratio of the number of labeled samples to the batch size is defined as γ. Then, the labeled training samples are defined as $\mathbf{X} = \{(x_i, v_i), i \in (0, 1, \ldots, \gamma B - 1)\}$, where x_i and v_i are the ith labeled sample and the label of the training data respectively; the unlabeled training samples are defined as $\mathbf{U} = \{u_i, i \in (0, 1, \ldots, (1-\gamma)B-1)\}$, where u_i is the ith unlabeled sample of the training data. The goal is to train a classifier f_θ with parameter θ on the dataset $\mathbf{X}$ and $\mathbf{U}$ to achieve the optimal performance on the entire dataset.

The framework of the semi-supervised view classification is shown in Fig. 5. The loss function designed for view classification consists of three terms: the supervised classification loss, the pseudo-label classification loss, and the consistency loss. The supervised classification loss is based on cross-entropy as

$$\mathcal{L}_{\text{sup}} = \frac{1}{\gamma B - 1} \sum_{i=0}^{\gamma B} H\left(v_i, p\left(y|x_i\right)\right) \tag{19}$$

where $H(\cdot)$ denotes the cross entropy function. Based on *FixMatch* [46], we mainly employed two SSL components: pseudo-labeling and consistency regularization. The pseudo-labeling method works as follows: input unlabeled data into the model, then filter the predicted probability outputs that meet the threshold conditions and convert them into pseudo one-hot labels. Subsequently, these pseudo-labels are utilized to further train the model

$$\mathcal{L}_p = \frac{1}{(1-\gamma)\, B} \sum_{i=0}^{(1-\gamma)B-1} \mathbb{1}_{\max(p_i) \geq \tau} H\left(T\left(p_i\right), p_i\right) \tag{20}$$

where $p_i = p(y|u_i)$, τ is the threshold, $T(\cdot)$ represents the transformation of a probability distribution into one-hot distribution; therefore, $T(p_i)$ is the pseudo-label of the unlabeled data u_i.

The consistency regularization method enhances the representational ability of the model by leveraging the perturbations introduced by data augmentation. This method assumes that

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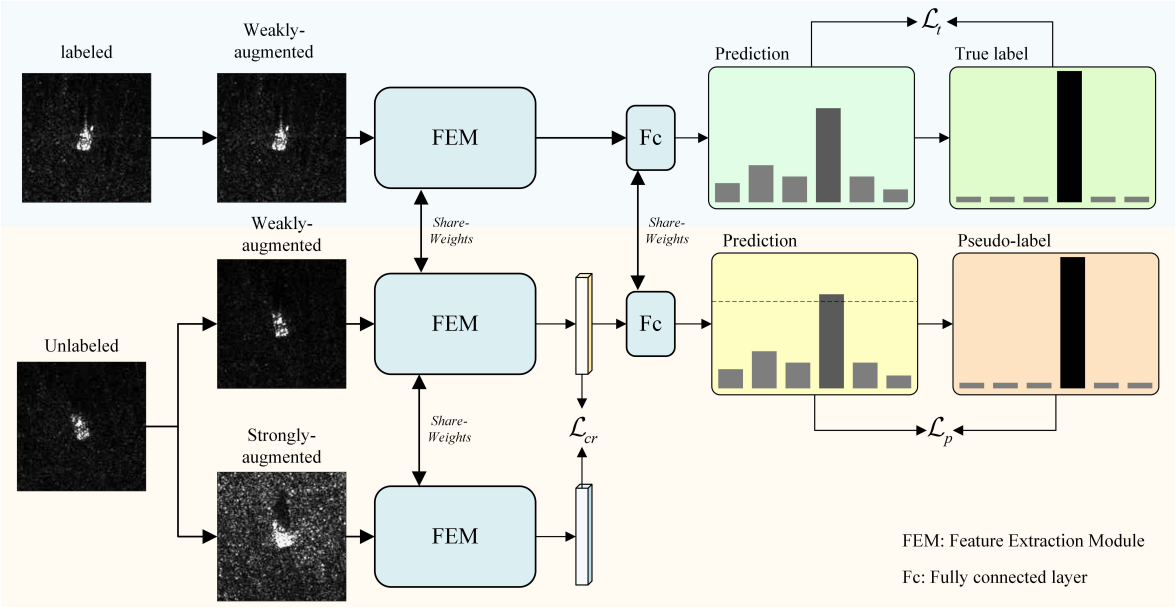

Fig. 5. Framework of the semi-supervised view classification. The labeled images are weakly augmented and then input into the classification network, and the cross-entropy loss is calculated normally. The unlabeled images are subjected to weak augmentation and strong augmentation, respectively, to obtain two versions. The consistency regularization loss is calculated on the output of the feature extraction module, and the cross-entropy loss using pseudo-labels is calculated on the output classification prediction probabilities.

when the same image is processed through different data augmentation methods, the output of the model should be similar. Different from *FixMatch*, we apply consistency regularization to constrain the feature representations rather than the probability outputs to improve the learning ability of view features. In terms of data augmentation, for SAR images, we mainly use various methods such as adaptive quantization [52], dilation and erosion, noise addition, geometric transformation, and filtering transformation for data augmentation. For optical images, the data augmentation methods are the same as those in *RandAugment* [53]. Consistency regularization is achieved by constructing a loss function that constrains the outputs of two augmented positive samples to be consistent, which can be expressed as

$$\mathcal{L}_c = \frac{1}{(1-\gamma)B} \sum_{i=0}^{(1-\gamma)B-1} \left\| 1 - \cos\left(g_\theta\left(\alpha_1\left(u_i\right)\right), g_\theta\left(\alpha_2\left(u_i\right)\right)\right)\right\|^2 \tag{21}$$

where $g_\theta(\cdot)$ represents the feature extractor with parameter θ. In this article, it is constructed by using ResNet50 followed by a fully connected layer, and the dimension of its output vector is D_g. $\alpha_1(\cdot)$ and $\alpha_2(\cdot)$ represent data augmentation processes.

Therefore, the total loss function designed for view classification is expressed as

$$\mathcal{L}_{\text{semi}} = \mathcal{L}_{\text{sup}} + \alpha_p \mathcal{L}_p + \alpha_c \mathcal{L}_c \tag{22}$$

where α_p and α_c are hyperparameters. Without loss of generality, we set $\alpha_p = 1$, $\alpha_c = 1$ in the experiments.

IV. Experiment

In this section, we first introduce the datasets used in the experiments and the experimental environment. Then, we present the ablation experiments and comparative experiments of the proposed multiview SAR ATR method based on the proposed *SeVi* model. Next, we describe the relevant experiments on applying the proposed incomplete multiview completion strategy to the *SeVi* model. Finally, we introduce the experiment of using the semi-supervised classification method to construct nonstrict multiview recognition datasets from existing SAR ATR datasets.

A. Dataset Description

We have collated or constructed three multiview SAR ATR datasets from existing SAR ATR datasets: 1) *MV-ATRNet-STAR:* a multiview SAR civilian vehicle target dataset built based on the ATRNet-STAR dataset [6]; 2) *MV-MSTAR:* a multiview SAR military vehicle target dataset constructed based on the MSTAR dataset [54]; and 3) *MV-SAR-Aircraft:* a multiview SAR aircraft target recognition dataset built based on the SAR-Aircraft-1.0 dataset [55]. In the experiments, we uniformly set the maximum number of views to 4 according to the model in Fig. 2. Typical multiview images of targets in these datasets are shown in Fig. 6.

The ATRNet-STAR dataset is an SAR vehicle recognition dataset containing 40 different categories of civilian vehicles collected under various real imaging conditions and scenarios, with over 190 000 well-annotated samples. The data was acquired by an airborne SAR platform, using the strip-map imaging mode, operating in X-band and Ku-band, with a resolution ranging from 0.12–0.15 m. The depression angles include 15°, 30°, 45°, and 60°, and multiple polarization forms are covered. The EOC-Azimuth experimental setting of this dataset compiles target data from different azimuth angles. The azimuth angles cover 0° to 360° with an interval of 5° and detailed annotations. Based on the aforementioned view classification model, we divided the images into four-view categories, including the forward and backward view v_0,

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TABLE I

STATISTICS ON THE NUMBER OF TARGETS IN THE MV-ATRNET-STAR DATASET

target type	Buick Excelle GT	Buick - GL8	CNHTC - HOWO	Chang'an CS75_Plus Plus	Chang'an Starlight 4500	Changfeng - CFA6473C	Changlin - Cheetah	Chery Arrizo 5 8228-5	Chery - qq3	Chevrolet Blazer 1998
target num	488	581	524	744	594	388	528	500	576	420
target type	Dongfeng Duolika	Dongfeng EQ6608LTV	Dongfeng Forthing Lingzhi	Dongfeng Tianjin DFH2200B	Dongfeng Tianjin KR230	FAW BJ1045-J6P	FAW Jiabao T51	Foton V9JB5-54	Great_Wall Voleex C50	Great_Wall poer
target num	464	732	672	528	524	524	468	568	753	637
target type	Hawtai - EV160B	Hongqi - CA7180A3E	Hongqi - h5	Huanghai - N1	Hyundai HLF25 II	JAC - Junling	JINBEI - SY5033XJH	Jeep - Patriot	Lincoln - MKC	Lveco Proud 2009
target num	636	597	492	511	528	585	672	484	524	648
target type	MAXUS - V80	Mitsubishi Outlander 2003	SDLG - ZL40F	SHACMAN DeLong M3000	SHACMAN DeLong X3000	WAW Aochi 1800	WAW Aochi Hongrui	Wuling Rongguang V	Yangzi - YZK6590XCA	Yutong - ZK6120HY1
target num	520	600	752	416	507	504	698	527	552	572

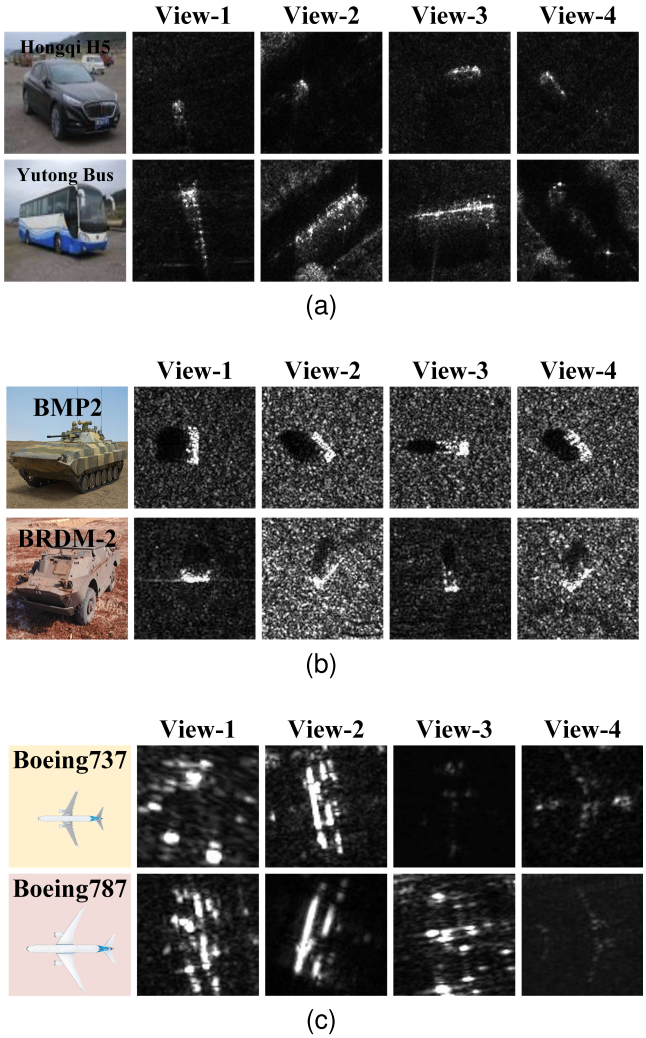

Fig. 6. Image samples of the multiview SAR target recognition datasets. Typical examples of two categories are provided for each dataset. (a) MV-ATRNet-STAR dataset. (b) MV-MSTAR dataset. (c) MV-SARAircraft dataset.

forward squint view v_1, side view v_2, and backward squint view v_3, and the angle ranges are as follows:

$$\begin{aligned} v_0 &\in (-22.5^\circ, 22.5^\circ) \cup (-180^\circ, -157.5^\circ) \cup (157.5^\circ, 180^\circ) \\ v_1 &\in (-67.5^\circ, -22.5^\circ) \cup (22.5^\circ, 67.5^\circ) \\ v_2 &\in (-112.5^\circ, -67.5^\circ) \cup (67.5^\circ, 112.5^\circ) \\ v_3 &\in (-157.5^\circ, -112.5^\circ) \cup (112.5^\circ, 157.5^\circ). \end{aligned} \tag{23}$$

TABLE II

STATISTICS ON THE NUMBER OF TARGETS IN THE MV-MSTAR DATASET

Type	BMP2	BTR70	T72	2S1	BRDM2
multi-view train	56	55	56	74	74
multi-view test	39	38	41	52	55
Type	**D7**	**BTR60**	**T62**	**ZIL131**	**ZSU234**
multi-view train	74	57	68	73	67
multi-view test	60	40	64	65	65

Table I presents the number of target samples in the MV-ATRNet-STAR data setup adopted for the multiview experiments. It is worth noting that the number of images in the multiview dataset we constructed is slightly less than that of the original dataset. This is because after performing view classification on the original dataset, the number of images in each view obtained is different. After we pair the multiview images according to the view with the smallest number of images, there are still some images remaining in other views. In practical tasks, the paired views and the remaining other views can be used to construct new samples, and even data augmentation can be adopted to further increase the amount of data to improve model performance. To avoid data leakage when dividing the training set, validation set, and test set in the experiment, we did not use more data augmentation methods in the multiview recognition experiment, but evaluated the model based on the target data given in the table. The same settings are adopted for the other two datasets.

The MSTAR dataset was collected and released by the Sandia National Laboratory in U.S. in the 1990s. This dataset contains SAR images of various targets and their variants under different scenes and observation conditions. The standard operating condition (SOC) of the dataset includes ten types of targets, and the collected azimuth angles cover 360°. The training data was collected at a depression angle of 17°, and the testing data was collected at a depression angle of 15°. Based on the view classification model, we divided the views of the original training data and test data, respectively. The number of targets in the training set and testing set of the MV-MSTAR dataset is shown in Table II. Fig. 6(a) shows the typical multiview images of the targets.

The MV-SAR-Aircraft dataset is constructed based on the SAR-Aircraft-1.0 dataset. SAR-Aircraft-1.0 consists of SAR images of three airports in Beijing, Shanghai, and Guangzhou, collected by the Gaofen-3 satellite, and provides target detection and recognition annotations for seven categories of

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TABLE III

STATISTICS ON THE NUMBER OF TARGETS IN THE MV-SAR-AIRCRAFT DATASET

Type	A220	A320	A330	ARJ21	B737	B787	other
num	600	300	50	200	500	500	600

civilian aircraft. The SAR images in the dataset have an imaging resolution of 1 m, with the spotlight imaging mode and one single polarization mode; the dataset contains a total of 16 463 aircraft target instances. First, we crop the target slices according to the ground truth labels and scale them to a size of 224×224. Second, for the images of each category of aircrafts, 100 groups of four-view label data were manually annotated. Then, the semi-supervised classification method is used to obtain the view labels of each sample in the complete dataset, which is discussed in detail in Section IV-D. Then, these are combined to form a multiview recognition dataset. The final number of targets was obtained after manually screening out some misclassified samples. Table III shows the statistical results of the number of targets in the MV-Aircraft dataset, and Fig. 6(c) shows the typical multiview images of the targets.

B. Multiview Recognition Experiment

In this part, we carry out multiview target recognition experiments on the three constructed datasets. For the proposed *SeVi* model, the default settings are as follows: the backbone network of the feature extraction module is ResNet50, the number of channels compressed by the linear mapping layer is 128, the feature embedding dimension is 768, the number of heads of the Transformer encoder is 12, and the depth is 4. In terms of optimization parameters, we choose the SGD optimizer, with a momentum parameter of 0.9, a learning rate of 1×10^{-4}, and a weight decay parameter of 1×10^{-4}. We set the batch size to 16 and the number of epochs to 200. On the platform with an Intel I9-13900 K CPU and a single Nvidia RTX4090 GPU, the average training duration is approximately 84 min.

In terms of the number of model parameters, under the default parameter conditions, the total number of model parameters is 85.93 M. Specifically, the number of parameters for a single feature extraction module is 23.44 M, and there are two such modules; the number of parameters for a single linear mapping module is 5.34 M, with a total of two modules; the number of parameters for the fusion module is 28.34 M, and the number of parameters for the classification head is 0.03 M.

1) Ablation Experiments: Based on the proposed SeVi model, we first conduct an ablation experiment on the number of views on the MV-ATRNet-STAR dataset to verify and analyze the fundamental advantages of target recognition using multiview feature fusion. We set different random seeds to perform multiple experiments, randomly selected and combined views, and used the average accuracy on the test set as the evaluation metric. For the single view, we use the fine-tuned ResNet50 model; for the multiple views, the main change is the number of embeddings in the *SeVi* model. Table IV shows

TABLE IV

ABLATION STUDY ON INFLUENCE OF NUMBER OF VIEWS ON MULTIVIEW RECOGNITION

view num	MV-ATRNet-STAR	MV-MSTAR	MV-SAR-Aircraft
1 view	92.81 ± 2.30	98.75 ± 0.32	89.56 ± 1.74
2 views	95.77 ± 1.42	98.91 ± 0.30	91.35 ± 3.46
3 views	98.12 ± 0.83	99.27 ± 0.16	91.92 ± 1.05
4 views	98.32 ± 0.80	99.28 ± 0.14	92.00 ± 0.74

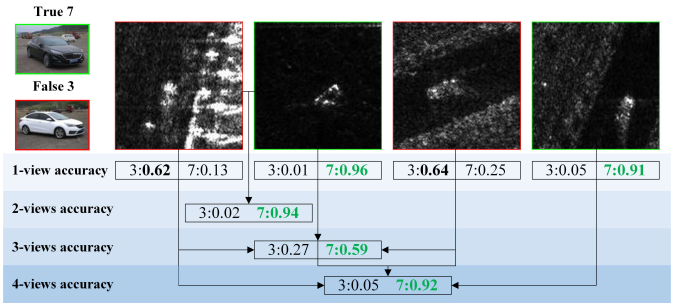

Fig. 7. Typical Example of target recognition with different numbers of views.

the mean, minimum, and maximum values of the average accuracy for different numbers of views on the three datasets.

The results show that increasing the number of views generally can improve the recognition accuracy to a certain extent, which proves that images from different views only contain partial target characteristics, and the model is able to learn the complementary features of different views. In addition, we also conduct cross-view verification, which involves training on one view and testing on another view. In this case, the recognition accuracy will decline significantly, indicating that there is a certain domain bias among images from different views. Fig. 7 presents the multiview recognition test results of the target Hongqi H5 (category 7) under models with different numbers of input views. The front-view and side-view scenarios are affected by strong background interference, which easily leads to confusion between the target and the Chery Arrizo 5 (category 3). In contrast, the target features under other views are relatively distinct, enabling the model to achieve correct recognition. It can be observed that multiview recognition has a strong ability to correct errors in misclassified views. We can assume that for a well-trained model, recognition errors occur because the model fails to capture meaningful features of the target, and the target feature vector falls on the boundary of multiple categories, while correct recognition is due to the model learning meaningful features and making credible judgments. Therefore, the fused features can still make correct judgments because certain views have learned the target information.

We employ three visualization methods to observe how the model learns multiview features. Specifically, we select two samples from the test set of MV-ATRNet-STAR. First, we output the feature maps of the semantic feature extraction module. Second, we obtain the locations of classification-critical regions focused on by the model using the Grad-CAM [56] method to generate heatmaps. Third, we calculate the attention scores between the view embeddings in the transformer fusion module and the class token via the attention

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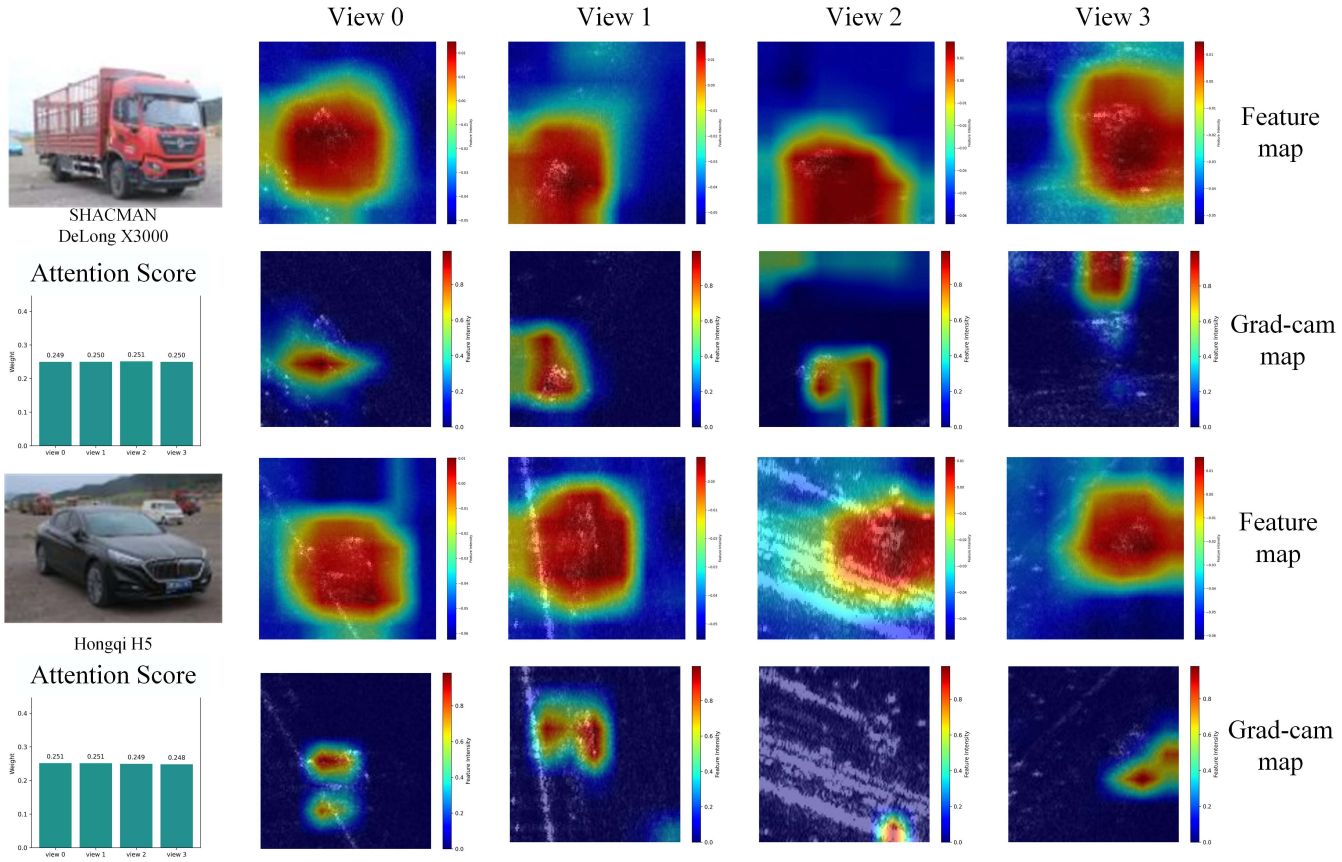

Fig. 8. Visualization results of feature learning in the multiview recognition model.

rollout [57] method, thereby deriving the fusion weight maps for each view. The results are shown in Fig. 8. As can be seen from the figure, the model can effectively focus on the important regions related to the target semantic categories in multiview images. Moreover, the model performs recognition by combining features from multiple views and exhibits a certain degree of anti-interference capability against views disturbed by complex backgrounds.

Regarding the multiview weights obtained by the attention rollout method, we find that when the training model is not converged, the weights fluctuate significantly, and there are obvious differences in the attention paid by the fusion module to different views. When the model is well-trained, the weights obtained by the attention rollout method are almost identical. We believe that the reason for this phenomenon is that the multiview features input to the transformer module are already extracted high-dimensional semantic features, which will not be reflected in low-dimensional information, such as image weights. When the model converges, the feature fusion module is capable of fully leveraging the features encapsulated in each view to determine the target category. In contrast to decision-level models, which tend to downplay or even disregard the contribution of images from a specific view, the proposed module avoids such suboptimal treatment of individual view information. In conclusion, the proposed model can effectively capture the feature information of targets from multiple views, thereby achieving reliable prediction results.

Then, we conduct ablation experiments on the key modules in *SeVi*, and the results are shown in Table V. The experimental dataset was MV-ATRNet-STAR, with the number of views fixed at 4 and other parameters remaining unchanged. "w/o view branch" refers to removing the entire branch of view feature extraction, feature embedding, and single-view view classification head from the original model, which is equivalent to "semantic branch only". Under this experimental setup, the model only includes the semantic branch and the backend Transformer-based feature fusion model, with its performance decreasing by 1.45%. This result indicates that using view features to perform positional embedding on semantic features can help the model capture and learn the relationships between views, thereby improving fusion recognition performance. "w/o single-view semantic classification head" means that the single-view semantic classification head is no longer used to assist the learning of the semantic feature extraction module during training. At this time, the model accuracy decreased by 8.03%, indicating that the single-view semantic classification head is quite important for the training of models with the CNN-Transformer architecture. In the "w/o single-view view classification head" setting, we removed the view classification head, used pretrained view classification weights, and froze the view branch. The model accuracy decreased slightly by 0.49% in this case, which can illustrate that in the original model, the multitask loss jointly optimizes the entire model, and the view branch also incorporates information from fusion

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classification. When there is no view pretraining, the model performance will drop to a level similar to that of the w/o view branch setting. The view classification head and view loss are still important because they enable the model to complete end-to-end view classification training while training the entire multiview recognition network.

Moreover, we conducted ablation experiments on the feature extraction networks. The ResNet18 and ResNet34 backbones have approximately 10.63 and 20.73 M parameters, respectively. For networks with the same structure, larger parameter counts correspond to stronger feature extraction capabilities. The optimized CSPDarkNet53 backbone, with around 26.58 M parameters, achieved higher performance. This indicates that selecting backbone networks with larger parameter counts can improve performance to a certain extent. Transformer-based fusion also showed better performance compared to ordinary fully connected layer fusion. In addition, parameter matching between the feature extraction module and the feature fusion module is also important. In experiments, using four-layer Transformer modules for feature fusion achieved a good balance between parameters and performance. However, excessively increasing the number of Transformer layers not only significantly increases model complexity but also easily leads to overfitting. In practice, adjusting the front-end and back-end parameter counts based on the task data can better achieve optimal feature extraction and fusion results simultaneously.

Furthermore, another important finding is that whether the feature extraction network adopts pretrained weights has a significant impact on model accuracy. In the experiment, the pretrained weights of the feature extraction module are obtained through training on the SARDet-100 K dataset [58]. When semantic and view pretrained weights were not used, the model performance dropped by more than 10%. We believe that the role of pretraining is to stabilize and accelerate model training, and generally, using pretrained models with strong representation capabilities can improve performance. However, the proposed CNN-Transformer model is more difficult to train due to issues such as multiple CNN branches and the lack of empirical bias in the Transformer. The training techniques for multiview recognition models with different architectures deserve detailed research and analysis.

2) Comparative Experiments: Furthermore, we conduct comparative experiments on the state-of-the-art methods for multiview recognition, and the results are shown in Table VI. Noted that due to the differences in experimental setups adopted in different SAR multiview recognition studies, we unified the data preprocessing methods, model backbones, training optimizers, etc. Moreover, we keep the basic architecture of various methods unchanged and remove some general attention modules. For methods based on the CNN architecture, ResNet50 was uniformly used as the feature extraction module or backbone network. First, we implement three basic fusion methods in [59]. Score fusion performs recognition on each view image first and then conducts decision fusion based on confidence; early fusion uses the first two blocks of ResNet50 to extract image features respectively, and then performs early feature fusion using channel concatenation and convolution layers; share-weight late fusion first extracts the embedding vector of each view, and then conducts channel fusion on the embedding vectors. The results show that share-weight late fusion has obvious performance advantages among the three basic fusion methods; the fusion of high-level features yields better results. Noted that in these three methods, multiple branches of the model all adopt shared parameters. Multiencoder late fusion is a feature fusion model with multiple independent encoders commonly used in multimodal tasks. Comparing the two late fusion methods, the share-weight late fusion has higher performance advantages with fewer parameters. The possible reasons for the poor performance of the multiencoder late fusion method could be that insufficient data volume in the SAR multiview recognition task leads to insufficient model representation learning; the increase in the number of parameters and branches of multiencoders makes training more difficult. Therefore, for SAR multiview recognition tasks with the current amount of data, it is necessary to cautiously adopt large-parameter models with multiple independent branches.

The adaptive weight fusion method [31] first uses a CNN to extract multiview features, then performs classification through a fully connected layer after summation and fusion. Its main feature is that a weighted branch is constructed based on the idea of self-attention for weighted summation. The weighted weights can be regarded as an indication of the contribution degree of the network to the fusion recognition of specific views. IA-AMF-Net [18] proposes a two-stage recognition method. First, a feature extraction module is trained to separately extract features from images of each view, and fusion is achieved by summing the feature vectors; then, a fused feature classification module is trained to obtain the final classification results. We implemented this by splitting the ResNet50 network into these two parts. This method can achieve accuracy close to the share-weight late fusion method and is capable of accepting inputs with a variable number of views. MIDFF [30] adopts a fusion approach that combines feature extraction with feature vector summation. It also employs an adversarial training method, where an image from other categories or views is input, and the contrastive loss is used to enlarge the gap between different classes. We reproduce this method by adding heterogeneous image input and adversarial loss to the IA-AMF-Net method. The results show that the adversarial loss can help feature learning and improve the recognition performance. The ResNet-LSTM [22] model employs LSTM to fuse the view image features extracted by CNN, and its fusion recognition performance is slightly lower than that of the share-weight late fusion model. This method is proposed for the scenario of SAR multiazimuth image sequence recognition. Logically, the input images need to satisfy the azimuth observation order; when this input requirement is not met, the advantage of LSTM in sequence modeling cannot be exerted, which may lead to limited recognition performance.

The ResNet-Transformer [26] model employs a Transformer to fuse the feature vectors of multiview images. In the experiment, the feature vectors obtained by ResNet50 are mapped into embeddings through a linear layer, and a small version ViT module is used to complete feature fusion and classification. It is found that the Transformer-based fusion

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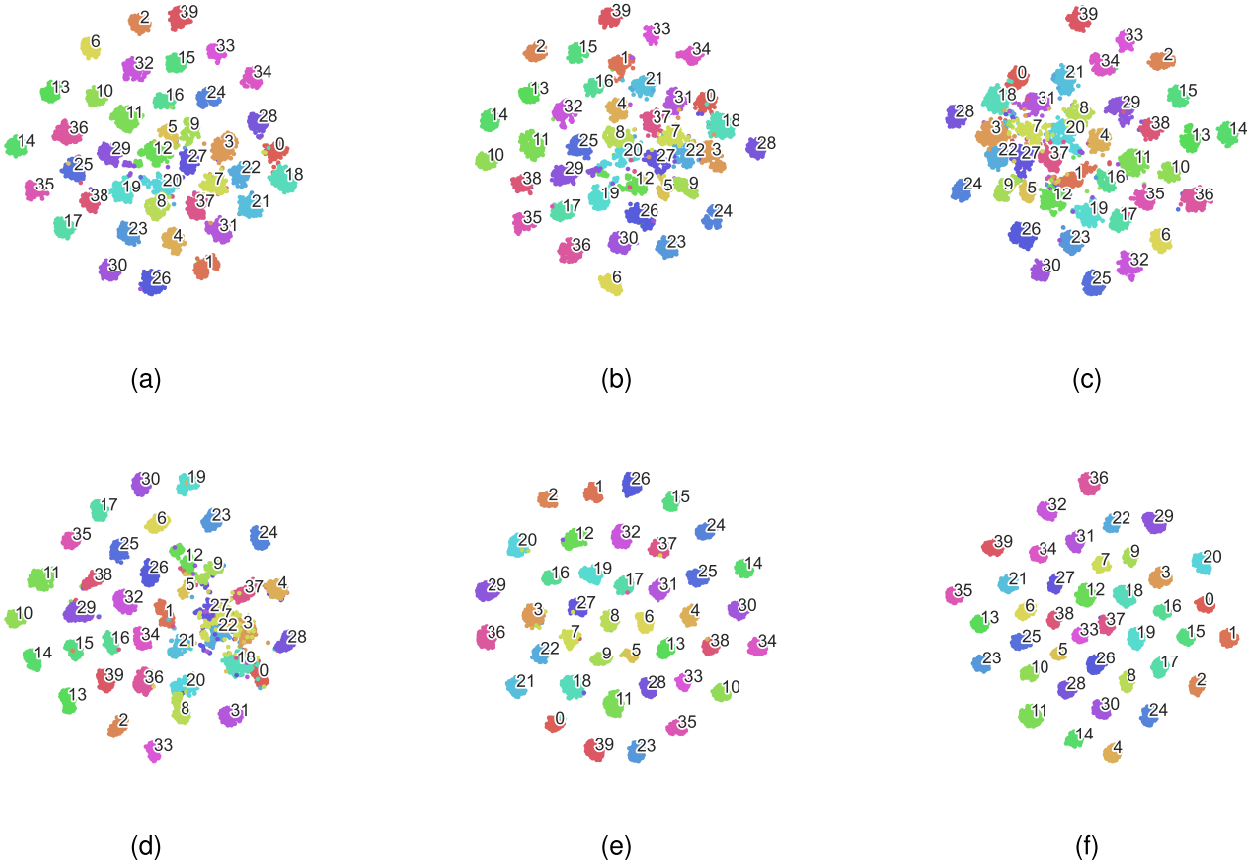

Fig. 9. Visualization results of multiview SAR target recognition model performance based on the t-SNE method. (a) Result of share-weight late fusion. (b) Result of MIDFF. (c) Result of ResNet-LSTM. (d) Result of ResNet-Transformer. (e) Result of ResNet Cross-attention Transformer. (f) Result of the proposed model *SeVi*.

architecture has a large number of parameters, leading to unstable model training and thus low final performance. In addition, the number of parameters of the feature extraction module and the feature fusion module in multiview fusion recognition needs to maintain a certain balance. The multiview model proposed in [16] achieves competitive performance by using a transformer module to calculate cross-attention between the main view and auxiliary views for view fusion, based on a basic CNN-based classification model. The proposed SeVi model follows the basic settings in the aforementioned experiments. Compared with other methods, the proposed model has certain advantages in both recognition accuracy and robustness.

We use t-SNE [60] to visualize the multiview SAR ATR recognition results of the models. The results are shown in Fig. 9. Visualization results of the performance of a multiview SAR target recognition model based on the t-SNE method. Each point represents a target sample containing four view images. It can be observed that the clustering results obtained by the proposed method exhibit compact intraclass distribution and significant interclass differences, indicating that the proposed model has effectively learned the target features and achieved high-precision recognition.

3) Further Experiments: Moreover, we analyze the effectiveness of view embedding. We select a variety of fusion methods for comparison with the method proposed in this article. On the MV-ATRNet-STAR dataset, the input order of the model is v_0, v_1, v_2, v_3. Fixed input views are used during the training process, and either fixed input views or disordered input views are used during inference. Disordered input views refer to the situation where the views of the images are unknown. The experimental results are shown in Table VII. It can be seen that both the FC fusion method and the LSTM fusion method have experienced significant performance degradation; the performance degradation of the Transformer method is relatively less than the former two, and the method proposed in this article has achieved the best performance. The experimental results show that view embedding can provide effective relative relationship information for the image features of different views to assist in fusion; the fusion

TABLE V
ABLATION STUDY ON KEY MODULES IN *SeVi*

Model	Avg.Acc	Δ
full model	**98.55 ± 0.39**	-
w/o view branch	97.10 ± 0.32	−1.45
w/o single-view semantic classification head	90.52 ± 2.40	−8.03
w/o single-view view classification head	98.06 ± 0.65	−0.49
w/ backbone ResNet18	91.98 ± 0.42	−6.57
w/ backbone ResNet34	96.14 ± 0.25	−2.41
w/ backbone CSPDarknet53	98.70 ± 0.42	+0.15
w/ fusion block fc	96.19 ± 0.18	−2.36
w/ fusion block Transformer	96.57 ± 0.42	−1.98

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TABLE VI
COMPARATIVE STUDY ON MULTIVIEW TARGET RECOGNITION METHODS

Method	MV-ATRNet-STAR	MV-MSTAR	MV-SAR-Aircraft
score fusion [59]	93.60 ± 0.45	98.19 ± 0.26	90.22 ± 1.10
early fusion [59]	79.81 ± 6.08	95.43 ± 1.43	80.95 ± 2.12
share-weight late fusion [59]	98.85 ± 0.19	99.95 ± 0.04	96.07 ± 0.27
multi-encoder late fusion [59]	85.10 ± 2.50	97.32 ± 0.98	83.69 ± 1.35
adaptive weight fusion [31]	96.41 ± 2.74	99.30 ± 0.34	94.75 ± 0.32
IA-AMF-Net [18]	97.29 ± 1.17	98.70 ± 0.16	94.17 ± 0.30
MIDFF [30]	97.90 ± 0.60	98.51 ± 0.14	94.86 ± 0.54
ResNet-LSTM [22]	95.73 ± 0.36	99.31 ± 0.34	94.84 ± 0.82
ResNet-Transformer [26]	94.55 ± 0.53	98.43 ± 0.56	90.90 ± 0.50
ResNet-Cross-attention Transformer [16]	99.14 ± 0.15	99.08 ± 0.22	95.93 ± 0.24
SeVi(ours)	**99.26 ± 0.13**	**99.95 ± 0.05**	**96.34 ± 0.35**

TABLE VII
EFFECTIVENESS ANALYSIS OF VIEW EMBEDDING

Method	fixed views	shuffled views
share-weight late fusion	97.09 ± 0.72	83.62 ± 2.17
ResNet-LSTM	95.45 ± 1.01	79.96 ± 1.98
ResNet-Transformer	89.67 ± 1.32	89.55 ± 1.56
SeVi(ours)	**98.34 ± 0.88**	**98.35 ± 0.69**

based on the self-attention mechanism has stronger robustness compared with other fusion methods.

C. Incomplete Multiview Recognition Experiment

The basic idea of incomplete multiview learning is to transform the incomplete multiview problem into a complete multiview problem through certain completion strategies and to reduce the impact of missing views on the remaining views. We conduct a horizontal comparison of five missing view completion methods on three datasets, using average accuracy as the evaluation criterion, and the results are shown in Table VIII. The monochrome method uses a solid gray image for completion; the Gaussian noise generates noisy images for completion; the random duplicate method randomly copies images from other existing multiviews for completion; the average embedding method simply calculates the mean of embedding vectors of a large number of samples for completion, without considering category differences. In the experiment, the *SeVi* model adopts the aforementioned basic parameters, with a circular queue length $N = 32$. We mainly set two typical conditions: 1) *Experimental Condition 1:* training was performed on the complete dataset, and in the test data, 50% of the samples randomly miss one view, and 50% randomly miss two views and 2) *Experimental Condition 2:* both the training data and test data have 50% of the samples randomly missing one view and 50% randomly missing two views. The results demonstrate that the proposed completion strategy maintained superior performance across all datasets. The semantic prototypes calculated by the proposed method are more stable, which helps the model fuse the incomplete multiview information.

Then we analyze the additional memory resources required by the proposed filling method. The dimension of the circular queue for each semantic category is $K \times D$, the vector length of the semantic clustering center is D, and the length of the queue pointer 1. The memory resources required for all N_s semantic categories are $N_s(\mathrm{KD} + D + 1)$. Similarly, the memory resources required for all N_v view categories are $N_v(\mathrm{KD} + D + 1)$. Under the default parameters, the increased memory is only 1.35 MB. We further carried out an analysis of the computational complexity. In each batch, the complexities of calculating the clustering centers of each semantic category and each perspective category are $O(N_s)$ and $O(N_v)$, respectively, and the complexities of enqueueing, dequeueing, and mean calculation of the circular queue are all $O(K)$. The actual comparison results are shown in Table IX, which demonstrate that the proposed method does not cause significant cost.

Finally, we conduct an experiment on the situations with more missing views on the SAR-ATRNet-STAR dataset. The samples in the training set and test set have the same conditions. The results are shown in Table X, which indicate that the proposed method has strong robustness and holds great promise in practical tasks.

D. Semi-Supervised View Classification Experiment

The semi-supervised view classification experiments are conducted on the MV-ATRNet-STAR and the MV-MSTAR datasets, which are equipped with complete view labels. We mainly explore how many manual labels are sufficient for the view classification task. In addition, we conduct a horizontal comparison of semi-supervised classification methods, which demonstrated the superiority of the proposed method.

According to the model established in Section III-D, we select ResNet50 as the backbone network. The proportion of labeled samples in each batch is set to $\gamma = 0.25$; the proportional parameters of the loss function are set to $\alpha_p = 1$, $\alpha_c = 1$; and the pseudo-label threshold is set to $\tau = 0.85$ on the MSTAR dataset and $\tau = 0.95$ on the self-built vehicle dataset, respectively. We adopt the SGD optimizer with the learning rate of 0.003, the momentum parameter $\beta = 0.99$; the batch size is 64, and the number of iterative updates is 2^{15}. As for the data augmentation strategy, we conduct a series of experiments on the data augmentation methods in RandAugment [53]. We find that three-channel color transformations have little effect on single-channel SAR amplitude images and finally select augmentation operations, including *random resize and crop*, *flip*, *rotate*, *cutout*, and *contrast* to suit the SAR modality. We choose the average accuracy of all categories as the evaluation criterion.

The results of the ablation experiment on the influence of the number of labeled samples on the accuracy are shown in Fig. 10. The experimental results regarding the impact of the number of perspective labels on the accuracy of

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TABLE VIII

COMPARATIVE STUDY ON FILLING STRATEGIES OF INCOMPLETE MULTIVIEW LEARNING

Condition	Method	MV-ATRNet-STAR	MV-MSTAR	MV-SAR-Aircraft
train: complete views test: 50% 1 missing view, 50% 2 missing view	monochrome	96.62	98.69	92.43
	gaussian noise	96.62	98.40	90.16
	random duplicate	97.05	98.75	90.16
	average embeddings	97.83	99.12	94.29
	proposed prototype	98.32	98.85	95.50
train: 50% 1 missing view, 50% 2 missing view test: 50% 1 missing view, 50% 2 missing view	monochrome	94.15	97.12	92.15
	gaussian noise	93.96	97.60	92.62
	random duplicate	95.32	97.08	92.77
	average embeddings	95.19	98.40	93.54
	proposed prototype	97.82	98.52	94.80

TABLE IX

COMPARISON ON CONSUMPTION OF SeVi UNDER COMPLETE MULTIVIEW AND INCOMPLETE MULTIVIEW CONDITIONS

Method	Memory Cost(MB)	Training Time(min)
SeVi with complete multi-view	329.05	229
SeVi with incomplete multi-view	330.40(+1.35)	237(+3.49%)

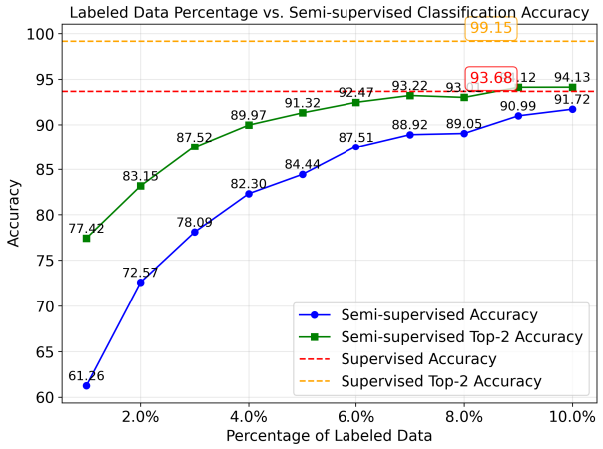
TABLE X

STUDY ON NUMBER OF MISSING SAMPLES

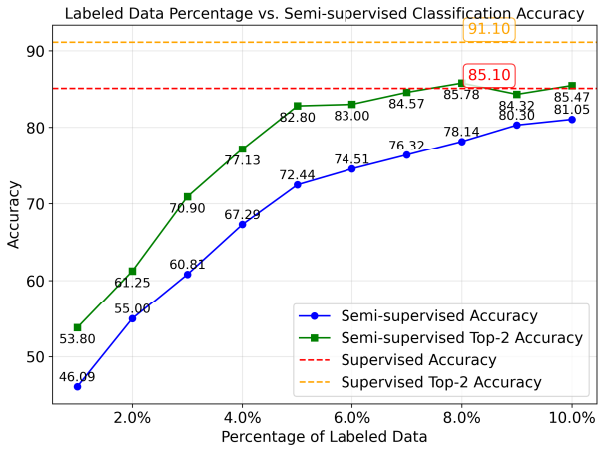
No.	Sample Condition	Accuracy
1	50% missing samples lack 1 view	97.24 ± 0.75
2	100% missing samples lack 1 view	97.12 ± 0.62
3	50% missing samples lack 1 view, and 50% missing samples lack 2 views	96.45 ± 1.02
4	100% missing samples lack 2 views	94.82 ± 0.86
5	50% missing samples lack 2 view, and 50% missing samples lack 3 views	92.73 ± 1.45

view classification are shown in the figure. We calculated the average accuracy and top-2 accuracy for label quantities ranging from 1% to 10%. On the MV-ATRNet-STAR dataset, using 10% of the labels achieved an accuracy of 86.72% and a top-2 accuracy of 94.13%, while the supervised method reached an accuracy of 93.68% and a top-2 accuracy of 99.15%. The difference between the classification accuracy using approximately 10% of the labels and the accuracy of fully supervised classification is within 5%. Most misclassified samples occur at the boundaries between two adjacent view categories. As indicated by the top-2 accuracy, the semi-supervised view classification can effectively learn features related to perspectives. The slightly lower overall accuracy on the MSTAR dataset may be attributed to imprecise manual annotations and a relatively small amount of data. It is worth noting that when experiments are conducted only on a dataset of a single category, a high classification level can be achieved with very few labels. However, in such cases, the model may only learn low-level features of the scene, leading to a serious overfitting problem and a lack of generalization ability on other categories. Fig. 11 shows the results of semi-supervised view classification using 10% of the labels. Overall, the discrete view classification model still exhibits well separability. Then, based on the obtained experience, we are able to carry out the semi-supervised view classification task and construct multiview recognition dataset on other larger SAR target recognition datasets.

We compare some typical unsupervised and semi-supervised classification methods, and the results are shown in Table XI. For the unsupervised methods, we have selected two clustering



(a)



(b)

Fig. 10. Impact of the number of view labels on classification performance. (a) Results on MV-ATRNet-STAR dataset. (b) Results on MV-MSTAR dataset.

algorithms, DeepClustering [43] and SCAN [61]. For the unsupervised methods, we conduct evaluations on the datasets formed by each category separately. We set the number of clustering centers to be equal to the number of views, select

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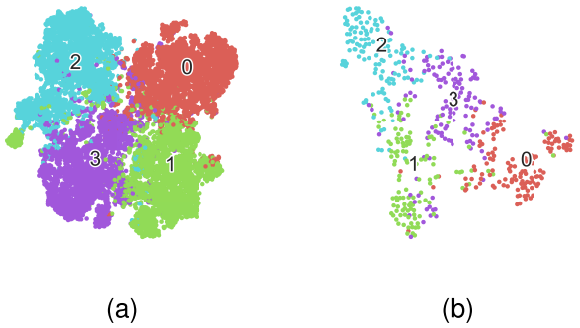

Fig. 11. Visualization of semi-supervised view classification obtained by t-SNE. (a) Visualization result on MV-ATRNet-STAR dataset. (b) Visualization result on the MV-MSTAR dataset.

TABLE XI
COMPARATIVE STUDY ON SEMI-SUPERVISED VIEW CLASSIFICATION

Method	MV-ATRNet-STAR	MV-MSTAR
DeepClustering (w/o label)	25.49	26.17
SCAN (w/o label)	26.81	27.24
PL	69.30	64.17
UDA	71.44	61.58
FixMatch	88.26	79.06
Ours	**89.56**	**83.45**
Fully-Supervised	93.08	86.27

the best match between the clustering results and the actual view labels as the view classification result for that semantic category, and finally calculate the average accuracy across all semantic categories. The results show that, due to the absence of prior information about views, it is difficult to converge toward the direction of view classification by merely utilizing the semantic information of the images. Pseudo-label (PL) and unsupervsied data augmentation (UDA) are both effective components of SSL. Compared with FixMatch [46], we have constrained the consistency loss to the visual representation and added a nonlinear projection head, which can improve the model performance to a certain extent. In addition, Zhang et al. [47] and Wang et al. [49] carried out adaptive optimization of the pseudo-label screening threshold based on FixMatch, and this can also be applied to our method to enhance its performance. We can conclude that for the task of SAR image view classification, semi-supervised classification is the optimal choice that comprehensively considers resource consumption and performance.

V. CONCLUSION

In this article, we designed a novel multiview feature embedding and fusion model (SeVi) assisted by view prior information. The model takes multiple view images of the same target as input and provides semantic classification predictions through feature-level fusion processing. Specifically, we adopt a semantic-view dual-branch design to obtain the semantic features and view features of each image respectively. Then, view embedding is used to add information from different views to semantic embedding, so as to obtain view-aware semantic embedding. Finally, the Transformer structure is utilized to realize feature fusion. We construct three multiview SAR ATR datasets based on existing SAR vehicle and aircraft ATR datasets for experimental verification. Experiments show that the proposed model can effectively improve the recognition performance.

From the perspective of improving the practicality of the proposed multiview SAR ATR model, we conduct further research on the problems of incomplete and scarce data in actual tasks. For the problem of incomplete multiview learning, we propose a prototype completion strategy, which enables the model to be applicable to complex and variable testing environments while maintaining high performance. Aiming at the problem of scarce data in existing multiview SAR recognition method, we propose to use a small amount of view annotations and semi-supervised classification to expand multiview target samples to support model development. In conclusion, the multiview SAR recognition framework proposed in this article can be better applied to actual complex scenarios. Multiview observation is one of the important development directions in the field of target recognition in the future. There are still many unsolved problems in multiview recognition under conditions such as few samples, domain adaptation, and multimodality. More research will be conducted on multiview feature learning in the future.

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